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VOCATIONAL EDUCATION AND TRAINING AUTHORITY

EDITED

CURRICULUM FOR INFORMATION COMMUNICATION TECHNOLOGY

PREPARED BY:

VETA HQ

BOX 2849

DAR ES SALAAM

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VETA VISION AND MISSION

VETA's Vision Statement:

"An excellent Vocational Education and Training (VET) system that is capable of supporting national social economic development in a global context".

VETA's mission statement

"To ensure provision of quality Vocational Education and Training that meets labour market needs, through effective regulation, coordination, financing and promotion, in collaboration with stakeholders".

ACKNOWLEDGEMENT

VETA wishes to express its deepest gratitude to all those who participated in the process of developing this curriculum. Gratitude is due to the experts for their significant contributions in the preparation of the curriculum.

Special thanks should go to the Trade Advisory Committee members for their valuable advices, inputs and ultimately their recommendations for the approval of this Curriculum.

Valuable inputs from employers in both formal and informal sectors during labour market surveys are gratefully acknowledged.

It is worth noting that, so many people were involved in the process whom we could not mention here, thus VETA would also like to express its sincere appreciation for their contributions in one way or another for the accomplishment of this work.

For and on behalf of:
VOCATIONAL EDUCATION AND TRAINING AUTHORITY

Leah D. Lukindo
DIRECTOR OF VOCATIONAL EDUCATION AND TRAINING

FOREWORD

Vocational Education and Training (VET) is a fundamental tool for socio-economic development of the nation. Due to its importance, the 1994 VET Act No. 1 was enacted, not only to ensure VET is properly financed, promoted and regulated, but also to set a legal framework within which vocational training should be provided.

The basic aim of providing VET is to impart vocational skills to learners, which match with career opportunities as supplied by the labour market. Preparing youngsters for the realities of earning a living is a responsibility shared by many different groups of people both inside and outside the education sector. All are aware that it is vital to ensure that learners gain the best possible academic or vocational qualifications, in order to provide them with a realistic chance of succeeding in today's highly competitive job market in both the formal and informal sectors of the economy. The quality of training provided should be judged above all by its impact- or potential impact - on practice. In this context, curriculum is viewed as a very significant instrument for guiding the entire educational and training process aimed at achieving relevance, credibility, coherence and portability of VET results. A good curriculum must support learners under the influence of a training institution, in achieving requisite learning experiences. It must focus on developing the whole person. Therefore, the VET curriculum integrates both technical and life skills to enable the learner understand other cross cutting issues. In developing this curriculum several factors were taken into consideration. Central to them were corporate elements such as VETA's vision and mission statements as well as technical aspects such as principles of learning and instruction, the role played by related subjects such as mathematics, communication skills and the like in facilitating the process of skills acquisition.

In order to meet rapid technological and socio-economic changes, VETA adopted competency-based curricula which are in a modular format and individually-paced with flexible entry and exit, skills and knowledge being broad-based to specialization. Collaboration with the labour market is seen to be of paramount importance because this collaboration guarantees a dynamic development and tailoring of vocational education and training in step with the rapidly changing qualification needs of trade and industry.

The process of developing this curriculum involved a range of experts from the field as well as practicing vocational teachers and educationalists under the guidance of curriculum development specialists.

Therefore the Board believes that this curriculum is learner focused, knowledge and experience based, labour market demands guided and hence it is able to contribute towards national vocational workforce development in the country.

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CHAIRPERSON VET BOARD **SECRETARY VET BOARD**

LIST OF ABBREVIATIONS

AIDS - Acquired immunodeficiency Syndrome

CBET - Competence Based Education and Training Approach.

HIV - Human immunodeficiency Virus

NVETB - National Vocational Education and Training Board

RVTSC - Regional Vocational training and Service Centre

VET - Vocational Education and Training

VETA - Vocational Education and Training Authority

CAD - Computer Aided Design

CA/CAD - Computer application with Computer Aided Design

ENG & COMM – English and Communication Skills

TD - Technical drawing

ES - Engineering Science

LS - Life Skill

EET - Entrepreneurship

NVA - National Vocation Award

CM - Computer Mathematics

GPA - Grade Point Average

ICT - Information and Communication Technology

LIST OF TAC MEMBERS, EXPERTS AND SECRETARIAT

TRADE ADVISORY COMMITTEE MEMBERS

1. Mr. Pius Wenga - Chairperson
2. Ms. Neema Mhondo - Member
3. Mrs. Anne Kilato - Member
4. Mr. Richard Wambali - Member
5. Mr. Stephen Mguto - Member
6. Mr. Nkundwe Moses Mwasaga- Member
7. Mrs. Leah D. Lukindo - Director of VET – Secretary

EXPERTS

1 Mr. Abdul A. Mollel - VETA HQ
2. Mr. Dennis M. Lupiana - IFM
3. Mr. Moses N. Mwasaga - DIT
4. Mr. Ulrick F. Mkenda - DIT
5. Mr. Joseph Mwakasege - VETA – CHANG'OMBE
6. Mr. Esrom Albano - RVTSC - MOSHI

SECRETARIAT

1. Mrs. Rehema M. Binamungu - Curriculum Development Specialist
2. Mrs. Stella M Ndimubenya - Curriculum Development Specialist

DEFINITION OF KEY TERMS

Competence: The ability to use knowledge, understanding, practical and thinking skills to perform effectively to the workplace standards required in employment.

Occupational Standards: Specific requirements of competences people are expected to demonstrate in a particular occupational area, including knowledge and relevant attitudes. They also act as performance tool of assessment of the pre – scribed outcomes.

Performance criteria: indicate the expected end results or outcome in form of evaluative statements.

Standard: it is a set of statement, which if proved true under working conditions, mean that an individual is meeting an expected level and type of performance

Assessment: The process of collecting evidence and making judgments on whether competency has been achieved, or whether specific skills and knowledge have been achieved that will lead to the attainment of competency.

Element: A sub- unit (step), which reflects learning sequence with the aim of achieving broad learning objectives of a unit.

Unit: A statement of broad learning objectives, which pre- scribe the requirements of a standard in form of practical skills, knowledge and appropriate attitudes

Underpinning Knowledge: This is essential knowledge needed in order to demonstrate competences that are associated in performing a given task (cannot be shown by performance)

Circumstantial knowledge: Detailed knowledge, which allows the decision-making in regard to different, circumstances and cross cutting issues.

1.0 INTRODUCTION

The Vocational Education and Training (VETA) is a body corporate established by the Vocational Education and Training Act No. 1 of 1994. VETA was established with an overall responsibility for ensuring the provision of required skills for effective socio economic development in both, the formal and informal sectors. The Act was passed with a focus on making the training system more flexible, cost effective and responsive to the demands of the labour market

In order to ensure that the flexibility of the training system is enhanced, VETA has adopted a Competence Based Education and Training (CBET) approach. The approach is flexible in responding to demands of the labour market and at the same time it puts emphasis on performance at the standards set by the labour market.

The implementation of the CBET approach becomes more effective if the curriculum content is reflecting the work place skills requirements. This pre - requisite has been adhered to in developing this curriculum as the process started by conducting labour market surveys. Furthermore, technical inputs were drawn from experts in the field, trainers, and curriculum development experts.

To ensure that vocational education and training contributes to learners' employability, personal and social development, the curriculum content has been designed in such a way that it covers the technical aspects in terms of the required skills and knowledge; as well as attitudinal aspects in relation to cross cutting aspects such as entrepreneurship, environmental and HIV/AIDS issues. These have been integrated so as to produce graduates who can face the challenges of the world of work in a holistic manner.

In this document important information concerning the curriculum for Information, has been provided. It includes general objectives and functions of VETA, the curriculum structure, learning and training methodologies, management of assessment certification and award system, required qualifications for vocational teachers to be eligible to teach the course, and monitoring and evaluation modalities. The course content is also provided.

2.0 OBJECTIVES AND FUNCTIONS OF VETA IN RELATION TO VET SYSTEM

(a) To establish a vocational and training system including both basic and specialised training to meet the needs of formal and informal sectors of the economy.

(b) To satisfy the demand of the labour market for employees with trade skills in order to improve production and productivity of the economy.

(c) To foster and promote entrepreneurial values and skills, as an integral part of all training programs.

(d) To promote on- the - job training in the industry for both apprenticeship training and for skills up – dating and upgrading.

(e) To raise the quality of vocational education and training being provided.

(f) To promote or provide vocational education and training according to needs, within the framework of overall national social – economic development plans and policies.

(g) To promote access to vocational education and training for disadvantaged groups.

(h) To promote and to provide tailor – made course programs and in – service training in order to improve the performance both of quality and productivity of the national economy.

(i) To provide a dual vocational education and training system, combining broad basic training, gradual specialisation and practical experiences from work, and

(j) To promote a flexible training approach and appropriate teaching methodologies.

3.0 CURRICULUM STRUCTURE

3.1 MEANING OF THE OCCUPATION

The occupation involves performing duties related to Information and Communication Technology activities. This includes: Managing Office application and Internet as well as maintaining computer hardware, computer software and network hardware.

General competencies of the occupation:

Competencies required from this occupation include the ability to:

- Use office application programs
- Design and implement websites
- Design graphics
- Maintain and repair microcomputers
- Maintaining of Operating System
- Maintain and repair networks
- Maintaining of Network Software
- Develop database application programs
- Performing Preventive Maintenance
- Writing Basic Program
- Managing Internet Services

Employment opportunities for the occupation:

The occupation entails many opportunity aspects:

a) Public sector:

Public training institutions, Municipals, Ministries, Departments, Agents and Councils.

b) Private sector

- Self-employment
- Computer workshops
- Computer retailers
- Internet café
- Stationeries
- Private training institutions
- Any organization with 5 or more computers

c) Non-Governmental Organization

3.2 AIM AND OBJECTIVES OF THE COURSE

AIM

The course aims at imparting the required skills, knowledge and attitudes to people who wish to work in the Information and Communication Technology industry.

OBJECTIVES

Trainee graduating from this course should be able to:

a) Prepare different documentations
b) Maintain microcomputers
c) Maintain software
d) Maintain network hardware
e) Develop websites
f) Develop different graphics
g) Create simple database application programs
h) Perform Preventive maintenance

3.3 CODING OF MODULES

LEVEL I

MODULE CODES	MODULES TITLES
ICT - 01 - 01	Maintaining Safety of Workshop and surrounding.
ICT - 01 - 02	Preforming Preventive Maintenances of Tools, Equipment & Computer
ICT – 01 - 03	Performing Electrical Joints.
ICT – 01 - 04	Building Simple Electric Circuits.
ICT – 01 - 05	Building Simple Electronic Circuits.
ICT – 01 - 06	Maintaining Computer and its Peripherals.
ICT – 01 - 07	Maintaining Operating System.
ICT – 01 - 08	Managing Word Processor and Spreadsheet Applications.

LEVEL II

MODULE CODES	MODULES TITLES
ICT – 02 - 01	Maintaining Network and its Components
ICT – 02 - 02	Maintaining Network Software
ICT – 02 - 03	Managing Database and Presentation Applications
FA 201	Field Attachment

LEVEL III

MODULE CODES	MODULES TITLES
ICT – 03 - 01	Manage Safe Work Environment
ICT – 03 – 02	Maintaining website
ICT – 03 – 03	Designing Graphics
ICT – 03 – 04	Managing Internet Services
ICT – 03 - 05	Basic Programming
ICT – 03 - 06	Managing Preventive Maintenances
ICT – 03 – 07	Managing Workshop Activities
FA 301	Field Attachment

3.4 SUPPORT SUBJECT

3.4.1 Related subjects

- Computer Mathematics
- English & Communication Skills
- Engineering Science
- Computer Aided Design
- Technical Drawing

3.4.2 Cross cutting skills

- Entrepreneurship Education and Training
- Life Skills.

3.5 ENTRY REQUIREMENTS

The minimum entry qualification for the course is:

- **Level I**

Primary school Standard VII certificate or;

Ordinary Secondary School Education or;

Apprentices from the industry with field experience in Information Technology for one year,

- **Level II**

The trainee should have achieved CBET performance grade C in continuous assessment in level I.

- **Level III**

A trainee should have a minimum CBET performance grade C in continuous assessment in level II or vocational certificate II.

3.6 STRUCTURE AND ORGANISATION OF THE COURSE CONTENT

The course content is presented in a matrix form whereby the following components are indicated.

1. Module number
2. Module title
3. Training duration of a module
4. Unit title
5. Performance criteria
6. Elements
7. Competence assessment
8. Training requirements

3.7 THE STRUCTURE OF CURRICULUM CONTENT ARRANGED IN SEMESTERS

LEVEL I

SEMESTER	CORE MODULE TITLE	HOURS	No OF PERIODS	MODULES TITLES FOR SUPPORT SUBJECTS	HRS	No OF PERIODS
1ST	**ICT-01-01** Maintaining safety of workshop and surrounding.	100Hrs	134	**CM** **01** Sets **02** Exponents Radicals and Logarithms	12Hrs 18Hrs	16 24
	ICT-01-06 Maintaining computer and its peripherals	190Hrs	253	**Eng & Comm** 1. **Grammar (1.1.-1.2)**	30Hrs	40
				TD **01.** Drawing plain geometry.	30Hrs	40
	ICT-01-08 Managing word processor and spreadsheet	100Hrs	134	**ES** **01** Basic concepts of engineering science. **02** Force in equilibrium.	18Hrs 12Hrs	24 16
				EET **01.** Entrepreneurship concept. **02.** Generating feasible business idea (unit 2.1)	18Hrs 12Hrs	24 16
				LS **01.** Understanding personalities. **02.** Good interpersonal relationship and effective communication **03.** Creative problem solving and effective decision making **04.** Negotiation and conflict resolution.	24Hrs 15Hrs 12Hrs 9Hrs	32 20 16 12

SEMESTER	CORE MODULE TITLE	HOURS	No OF PERIODS	MODULES TITLES FOR SUPPORT SUBJECTS	HRS	No OF PERIODS
2ND	**ICT-01-02** Perform preventive maintenance	80Hrs	107	**Eng & Comm**		
				01. Grammar (Unit 1.3)	24Hrs	32
	ICT-01-03 Perform electrical joint	20Hrs	27	**02.** Word formation	6Hrs	8
	ICT-01-04 Building simple electric circuits	20Hrs	27	**TD** **02.** Scale drawing	3Hrs	4
				03. Construction of pictorial drawing	27Hrs	36
	ICT-01-05 Building simple electronic circuits	80Hrs	107	**ES** **01.** Dynamics	24Hrs	32
				02. Determination of pressure	18Hrs	24
	ICT-01-07 Maintaining operating system	190Hrs	253	**EET** **02.** Generating feasible business idea (unit 2.2)	10.5Hrs	14Hrs
				03. Starting a business	19.5Hrs	26Hrs
				LS **03.** Sexual and reproductive health	27Hrs	36
				04. Gender concern.	6Hrs	8
				05. Achieving career goals and vision.	3Hrs	4
				06. Creative and critical thinking.	9Hrs	12
				07. Referrals and linkages.	3Hrs	4
				08. Customer care.	12Hrs	16
				CM **02.** Algebra	8Hrs	11
				03. Linear Programming	10Hrs	13

LEVEL I: SUMMARY DISTRIBUTION OF TIME

SEMISTER 1	CORE SUBJECTS	SUPPORT SUBJECTS	TOTAL
Number of hours	390	210	600
Number of periods (1 period = 45 Minutes	520	280	800
Percentage	65%	35%	100
SEMISTER 2	**CORE SUBJECTS**	**SUPPORT SUBJECTS**	**TOTAL**
Number of hours	390	210	600
Number of periods (1 period = 45 Minutes	520	280	800
Percentage	65%	35%	100

LEVEL II

SEMESTER	CORE MODULE TITLE	HOURS	NUMBER OF PERIODS	MODULES TITLES FOR SUPPORT SUBJECTS	HRS	No OF PERIODS
1ST	**ICT-02-01** Maintaining network and its components	141hrs	188	**CM** **05.** Planning and Elevations	17Hrs	23
				06. Trigonometric	17Hrs	23
	ICT-02-02 Maintaining network software	249hrs	332	**Eng & Comm** **03.** Conversation	30Hrs	40
				TD **04.** Construction of orthographic projection.	15Hrs	20
				05. Construction of sectional view	15Hrs	20
				ES **03.** Simple machines	9Hrs	12
				04. Heat	21Hrs	28
				05. Strength of materials	7.5Hrs	10
				06. Work energy and power	12Hrs	16

SEMESTER	CORE MODULE TITLE	HOURS	NUMBER OF PERIODS	MODULES TITLES FOR SUPPORT SUBJECTS	HRS	No OF PERIODS
				07. Friction	10.5Hrs	14
				EET **04.** Managing business (Unit 4.1 -4.4)	30Hrs	40
				CA **01.** Introduction to computer **02.** Office application (Unit 2.1 – 2.2)	3Hrs 27Hrs	4 36
2nd	**ICT-02-03** Managing database and presentations application	180Hrs	240	**EET** **03.** Managing business (unit 4.5) **04.** Getting into business	6Hrs 24Hrs	8 32
				CM **07.** Number system	14Hrs	19
				Eng & Comm **05.** Communication concept **06.** Applying writing and reading skills	12Hrs 18Hrs	16 24
				TD **06.** Drawing Descriptive geometry and auxiliary views **07.** Drawing of similar and equivalent areas **08.** Drawing Loci (Unit 8.1)	18Hrs 6Hrs 6Hrs	24 6 6
				ES **08.** Electricity and magnetism	30Hrs	40
				CAD **02.** Office application (2.3 – 2.4)	30Hrs	40

LEVELII: SUMMARY DISTRIBUTION OF TIME

SEMISTER 1	CORE SUBJECTS	SUPPORT SUBJECTS	TOTAL
Number of hours	390	210	600
Number of periods (1 period = 45 Minutes	520	280	800
Percentage	65%	35%	100
SEMISTER II	**CORE SUBJECTS +FA**	**SUPPORT SUBJECTS**	**TOTAL**
Number of hours	180 +240= 420	180	600
Number of periods (1 period = 45 Minutes	240 + FA	240	480
Percentage	70%	30%	100

LEVEL III

SEMESTER	CORE MODULE TITLE	HOURS	NUMBER OF PERIODS	MODULES TITLES FOR SUPPORT SUBJECTS	HRS	No OF PERIODS
1ST	**ICT-03-01** Manage safe work Environment **ICT-03-02** Maintaining website **ICT-03-03** Designing graphics **ICT-03-04** Managing internet services	130Hrs 130Hrs 150Hrs 80Hrs	173 173 200 107	**CM** **08** Calculating Devices **09.** Logic	8Hrs 9Hrs	11 12
				ENG & COMM **07.** Speaking and listening skills	30Hrs	40
				TD **08.** Drawing Loci (Unit 8.2) **09.** Construction of development and Interpretation **10.** Working drawing	6Hrs 15Hrs 9Hrs	8 20 12
	FA 301 **ICT-03-05** Basic programming	240Hrs 120Hrs	NA 160	**CM** **10.** Differentiations **11.** Solve equations	10Hrs 15Hrs	14 20

SEMESTER	CORE MODULE TITLE	HOURS	NUMBER OF PERIODS	MODULES TITLES FOR SUPPORT SUBJECTS	HRS	No OF PERIODS
2[ND]	**ICT-03-06** Managing preventive Maintenances **ICT-03-07** Managing workshop Activities	40Hrs 80Hrs	53 107	**CAD 03.** Computer Aided Design	60Hrs	80

LEVEL III: SUMMARY DISTRIBUTION OF TIME

SEMISTER 1	CORE SUBJECTS	SUPPORT SUBJECTS	TOTAL
Number of hours	480	120	600
Number of periods (1 period = 45 Minutes	640	160	800
Percentage	80%	20%	100%
SEMISTER II	**CORE SUBJECTS +FA**	**SUPPORT SUBJECTS**	**TOTAL**
Number of hours	240 +240= 480	120	600
Number of periods (1 period = 45 Minutes	320 + 320=640	160	800
Percentage	80%	20%	100%

4.0 LEARNING AND TRAINING METHODOLOGY

Learning and training methodology in CBET is based upon philosophy of learning through application, and learners' competent performance focuses on three domains i.e. cognitive, affective and psychomotor skills. Therefore, a variety of learning and teaching methods will be applied in delivering the curricular content. Normally, the vocational teacher functions as a facilitator hence learning and training is learner centred. Learning and training however will include institutional and industrial based training (field attachment). Field attachment will be part of training. Its assessment modalities will be managed by using a logbook comprising occupational areas to be covered.

4.1 FIELD ATTACHMENT

The period of field attachment is 8 (eight) weeks for level II and Level III, during which the vocational teachers should visit the trainees at least once to ascertain the learning and training performances.

There should be a logbook for each trainee suggesting areas to be covered and the actual areas covered. The industry to which the trainee is attached shall provide a supervisor to guide and assess him/her in the day-to-day activities.

5.0 MANAGEMENT OF ASSESSMENT

In a competence – based assessment system, the purpose of assessment is to collect sufficient evidence that the individuals can perform to specified standards in a specific role.

5.1 MODES OF ASSESSMENT

In implementing CBET, testing and assessment shall be conducted in two ways; internal and external assessment according to assessment guidelines.

Internal Assessment

This is a continuous or formative assessment which is conducted by a vocational teacher. Each unit has to be assessed and the trainee has to meet a required standard before she/he can proceed to the next unit.

Final Assessment

Final assessment is a quality assurance process conducted by a licensed external assessor before the learner is awarded a certificate of competence.

Validation of Assessment

Validation of final assessment is conducted by licensed experts from the industry in the relevant field with the objective of validating final assessment and the process of learning.

5.2 FINAL ASSESSMENT PLAN

Due to semesterization some subjects will be completed before the end of the training circle hence the need to preparation of this plan**.**

LEVEL	**SUBJECT(S)**
I	Life skills – All modules
II	1. Entrepreneurship Education and Training (EET) – All modules
	2. Engineering Science (ES) – All modules
	3. Computer Mathematics (CM) – MODULE 1-7
	4. Technical Drawing (TD) – MODULE 1- 8 (Unit 8.1)
	5. English and communication skills (Eng & Comm) – MODULE 1- 5
	6. Computer Application with CAD – MODULE 1-2
	7. CORE SUBJECTS – MODULE: All modules up to level II
III	1. Computer Mathematics (CM) – Module 8 -11
	2. Technical Drawing (TD) – MODULE 8 (Unit 8.2) – 10
	3. English and communication skills (Eng & Comm) – MODULE 6
	4. Computer Application with CAD – MODULE 3 - 4
	5. CORE SUBJECTS (ICT) – MODULE: All modules in level III

5.3 ROLES OF A VOCATIONAL TEACHER IN ASSESSMENT

In CBET, the teacher is an assessor in continuous assessment and responsible for completing the log sheets and transfer the grade attained by a trainee to a log book. The vocational teacher therefore should make sure that the log book is endorsed accordingly.

5.4 ROLES OF A TRAINEE IN ASSESSMENT

The trainee is responsible for identifying her/his own strengths and weaknesses i.e. she/he will conduct self-assessment under the guidance of a teacher or assessor. The trainee is also responsible for endorsing the results for the continuous and final assessment.

5.5 ASSESSMENT RECORDS

Progress reports for every individual trainee should be recorded in the log sheets and logbook as evidence that will be used to support claims of achievement. The vocational teacher has a duty to keep and maintain records of a trainee.

6.0 CERTIFICATION AND AWARD

6.1 Award and Certification - Description

a) The VET system covers the initial three levels (Level I –III) out of the 8 levels in the TVET Qualifications Framework.
b) The award in VET system is known as National Vocational Award (NVA).
c) To qualify for the NVA, one needs to have the required competencies in the three domains of learning i.e. Psychomotor (Practical skills), Cognitive (theory) and Affective (attitudes) domains in a relevant field.
d) NVA level I and II are voluntary exit points.
e) NVA Level III is a compulsory exit whereby one is regarded to have completed vocational education and training.

Level qualifications, qualification award, award requirements and competence descriptors are as hereunder:-

QUALIFICATION LEVEL	QUALIFICATION AWARD	AWARD REQUIREMENTS	COMPETENCE DESCRIPTOR
NVA Level III	Vocational Certificate III	Pass in advanced core subjects and support subjects	The holder of the qualification will be able to apply Advanced vocational skills and knowledge (be able to perform abstract tasks and supervise subordinates).
NVA Level II	Vocational Certificate II	Pass in core subjects (theory & practical), field attachment and all supporting subjects.	The holder of the certificate will be able to perform routine and non routine tasks including a limited range of abstract.
	Vocational Certificate I	Pass in core subjects (theory & practical) and field attachment	The holder of the certificate will be able to perform routine tasks.
NVA Level I	No qualification award. A Transcript will be issued by Zonal Offices	Pass in core prescribed modules through continuous assessment	The holder of the transcript will be able to perform routine tasks.

6.2 Weight and Contributions

a) Continuous assessments in all subjects shall carry 60% and final assessment shall carry 40%);

b) For **support subjects** with practical and theory components, the practical component shall carry 60% and the theory component shall carry 40%;

c) Field Attachment is a prerequisite for taking final examinations and and award;
d) Final assessment shall carry 40% in all subjects components i.e theory, practical(project work) and support subjects;
e) **Project work** shall be the final assessment for core subject practical component; and
f) Grade point will be calculated to determine a candidate's Grade Point Average (GPA) which will further determine ones overall grade

6.3 Pass Mark

a) Pass mark for Vocational certificate 1,Vocational certificate II and Vocational Certificates III in all subjects/components shall be 60% this applies for continous and final assessments; and
b) Field Attachment, whose minimum pass mark is 90%, is compulsory for every learner and it will be a standalone component;

6.4 NVA Grading System

The emphasis in CBET is on performing to the required standards in workplaces. Grades for each NVA Module and hence entire Level shall be assigned with reference to the grades meanings/definitions provided in Table 1 and the ranges of scores for the various NVA levels are provided in Table 2.

Table 1: Definitions of Letter Grades corresponding Grade Points

NVA LEVEL 1-3		
Grade	**Description**	**Grade Point**
A	**Excellent:** Independent, efficient, accurate work of extra ordinary outstanding quality	4.0
B	**Very Good:** Independent, efficient, accurate work of acceptable quality	3.0
C	**Good:** work of acceptable quality	2.0

NVA LEVEL 1-3		
Grade	**Description**	**Grade Point**
D	**Poor:** falls short of most of the critical competencies	1.0
F	**Very poor:** Falls short of all critical competencies	0.0
I	Incomplete	0.0
Q	Disqualified	0.0

NB: Candidates with grade A, B and C will qualify for the NVA.

Table 2: Ranges of scores for different grades

Ranges for NVA levels and relevant GPA are provided as follows:

Grade		Score Range	GPA
A	Excellent	90- 100	3.5 - 4.0
B	Very Good	75 - 89	3.0 - 3.4
C	Good	60 - 74	2.0 - 2.9
D	Poor	45 - 59	1.0 - 1.9
F	Very poor	00-44	
I		Incomplete	
S		Supplement	
Q		Disqualified	

Candidates with grade A, B and C will qualify for the NVA.

7.0. VOCATIONAL TEACHERS' QUALIFICATIONS

The minimum qualifications of a Vocational Teacher to conduct this course are:

a) A holder of Level III in ICT or equivalent with teaching certificate plus 3 years work experience is eligible to teach level I – II and can assist in practical sessions for Level III programme.
b) Diploma / Degree in Computer Science/Computer Engineering/IT
c) Vocational Teaching certificate from a recognised institution.
d) Working experience in the related field of not less than 2 years.

8.0. MONITORING AND EVALUATION

Implementation of the curriculum will be continuously monitored and summatively evaluated i.e. At the end of the course.

9.0 REFERENCE BOOKS

9.1 REQUIRED BOOKS

1. Michael Miller (2003): Absolute Beginner's Guide to Upgrading and Fixing Your PC, Que
2. Lemay, L (1999): Teach Yourself Web Publishing with HTML 3.2 in 14 days, Sams Net, Indianapolis.
3. Gary Cornell (1998): Visual Basic 6 from the Ground Up by McGraw Hill
4. Joe Habraken (2003): Absolute Beginner's Guide to Networking, Fourth Edition, Que
5. Shelley O'Hara (2005): Absolute Beginner's Guide to Microsoft® Windows® XP, Second Edition, Que

9.2 RECOMMENDED BOOKS

1. Jim Boyce (2003): **Absolute Beginner's Guide to Microsoft® Office 2003**, Que.
2. Michael Miller (2005): **Absolute Beginner's Guide to Computer Basics**, THIRD Edition, Que.
3. By Greg Perry (1999): **Teach Yourself Visual Basic 6 in 21 Days**: Complete Training Kit Sams.
4. Adobe Creative Team (1997) **Adobe PageMaker 6.5 Classroom In A Book**, Adobe Press.
5. Lisa Matthews (2003): **25 Things to Make and Do in Adobe**® Photoshop® Elements® 4.
6. John Levine (2000): **Internet Secrets** 2nd Ed., Hungry Minds.
7. Adrian Wong (2004); **Breaking Through the BIOS Barrier**: The Definitive BIOS Optimization Guide for PCs, Prentice Hall.
8. Heathcote, P. M (2000): **'AS' Level ICT**, Payne-Gallway Publishers Ltd.
9. White, R (1998): **How Computers Work**, 4th Edition, Macmillan Publishing, Indianapolis.

10. Shelley, J and Hunt, R (1984): **Computer Studies A First Course**, 2nd Edition, Pitman Publishing.
11. MacRae, K (2001): **The Do-It-Yourself PC Book**, Osborne/McGraw-Hill.

10.0 COURSE CONTENT OUTLINES FOR INFORMATION AND COMMUNICATION TECHNOLOGY

LEVEL: I

MODULE: 1 MODULE TITLE: MAINTAINING SAFETY OF WORKSHOP AND SURROUNDING [DURATION 100 HRS]

UNIT NO. 1 UNIT TITLE: MAINTAIN WORKSHOP SAFETY Duration: 15Hrs

PERFORMANCE CRITERIA	ELEMENTS	PROCESS ASSESSMENT	COMPETENCE ASSESSMENT		
			Service ASSESSMENT	**UNDERPINNING KNOWLEDGE ASSESSMENT**	**TRAINING REQUIREMENTS**
A person who has achieved this unit should be able to maintain workshop safety as per workshop rules and regulations.	1.1. Maintain workshop rule. 1.2. Maintain workshop working environment 1.3. Maintain personal safety	**The trainee should be able to:** • Select tools, equipment and safety gears. • Identify types of fire. • Operate firefighting equipment. • Observe workshop cleanliness. • Observe workshop safety regulations. • Store tools and equipment.	Workshop safety maintained according to rules and regulations.	**Detailed Knowledge of:** **Method used:** **Detailed knowledge of:** **Method used:** The trainee should explain different ways of maintaining workshop safety. **Principles:** The trainee should explain the principles of: • Firefighting. • General cleaning. **Theories:** The trainee should explain: • Workshop rules and regulations. • Different classes of fire. • Use of fire alarms and detectors. • Firefighting equipment and appliances.	This unit can be achieved at a work place or training institution. The following tools, equipment and safety gears should be available:- • Firefighting equipment. • Mark postures. • Dust bins. • Cleaning Agents • Mop. • Safety clear glasses. • Over-coat • Gloves. • Dusk Mask. • Computer Tool Kit.

PERFORMANCE CRITERIA	ELEMENTS	PROCESS ASSESSMENT	COMPETENCE ASSESSMENT		
			Service ASSESSMENT	UNDERPINNING KNOWLEDGE ASSESSMENT	TRAINING REQUIREMENTS
				• Personal protective equipment. **Circumstantial knowledge** **Detailed knowledge about:** • Workshop safety precautions. • Safe handling of tools and equipment. • Waste disposal.	

LEVEL: I

MODULE NUMBER: 1 **MODULE TITLE: MAINTAINING SAFETY OF WORKSHOP AND SURROUNDING**

UNIT NO: 2 **UNIT TITLE: HANDLING HAZARDS** **DURATION: 45 HRS**

PERFORMANCE CRITERIA	ELEMENTS	COMPETENCE ASSESSMENT			TRAINING REQUIREMENTS
		PROCESS ASSESSMENT	SERVICE ASSESSMENT	UNDERPINNING KNOWLEDGE ASSESSMENT	
A person who has achieved this unit should be able to handle hazards as per occupational health and safety procedures.	1. Handling mechanical hazards. 2. Handling physical hazards. 3. Handling chemical hazards. 4. Handling biological hazards. 5. Handling ergonomics.	**The trainee should be able to:** • Select relevant Computer safety gears • Interpret Computer safety rules and regulations • Identify classes of hazards • Identify hazardous materials • Observe safety precautions while handling hazards • Use all of Computer safety equipment • Handle different classes of hazards • Clean tools, equipment and workplace	Hazards handled as per occupational health and safety procedures.	**Knowledge evidence:** **Detailed knowledge of:** **Method used:** The trainee should explain how to:- • Identify potential hazards • Handle potential hazards • Use all Computer safety equipment **Principles:** The trainee should explain principles of:- • Identifying potential hazards. • Handling potential hazards. • Using safety equipment.	This unit can be achieved at a work place or in training institution. The following equipment, tools and safety gears should be available:- • Safety equipment • Equipment manuals • Incinerators • Dust bins • Gloves • Masks • Dust coat • Safety boots • Safety goggles • Ear muffs

PERFORMANCE CRITERIA	ELEMENTS	COMPETENCE ASSESSMENT			TRAINING REQUIREMENTS
		PROCESS ASSESSMENT	**SERVICE ASSESSMENT**	**UNDERPINNING KNOWLEDGE ASSESSMENT**	
		• Store tools and equipment		**Theories:** The trainee should explain:- • The different classes of hazards • Effects of potential hazards • The ways of handling differential potential hazards **Circumstantial knowledge: Detailed knowledge about** • Safety precautions while handling potential hazards. • Safe handling of computer safety equipment	

LEVEL: I

MODULE NUMBER: 1 **MODULE TITLE: MAINTAINING SAFETY OF WORKSHOP AND SURROUNDING**

UNIT NO: 3 **UNIT TITLE: HANDLING FIRE ACCIDENTS** **DURATION: 15 HRS**

PERFORMANCE CRITERIA	ELEMENTS	COMPETENCE ASSESSMENT			TRAINING REQUIREMENTS
		PROCESS ASSESSMENT	SERVICE ASSESSMENT	UNDERPINNING KNOWLEDGE ASSESSMENT	
A person who has achieved this unit should be able to handle fire accidents as per rules and regulation.	1. Handling different classes of fire. 2. Handling firefighting equipment and materials.	**The trainee should be able to:** • Select relevant fire protective gears • Identify firefighting equipment • Use fire protective gears • Identify classes of fire • Identify types of fire extinguishers • Apply Firefighting rules and regulations • Apply right class of fire extinguisher • Observe safety precautions while handling fire accidents • Handle Firefighting equipment	Fire accidents handled as per rules and regulation.	**Knowledge evidence:** **Detailed knowledge of:** **Method used:** The trainee should explain how to: • Apply Firefighting rules and regulations • Identify different types of fire extinguisher • Apply the right type of fire extinguishers • Apply right type of Firefighting materials **Principles:** The trainee should explain principles of: • Identifying different types of fire extinguishers • Checking and testing fire extinguishers	This unit can be achieved at a work place or in training institution. The following equipment, tools and personal protective gears should be available:- • Firefighting equipment. • First aid kit • Fire jackets. • Fire blankets. • Heat resistive gloves. • Fire goggles. • Fire gas masks. • Fire helmets. • Fire safety boots.

PERFORMANCE CRITERIA	ELEMENTS	COMPETENCE ASSESSMENT			TRAINING REQUIREMENTS
		PROCESS ASSESSMENT	**SERVICE ASSESSMENT**	**UNDERPINNING KNOWLEDGE ASSESSMENT**	
		• Handle Firefighting materials • Handle different fire classes • Clean tools, equipment and work area. • Store tools and equipment		• Applying right class of fire extinguishers. **Theories:** The trainee should explain:- • Importance of handling fire accidents • Common classes of fire • Handle different types of fire • Importance of checking and servicing fire extinguishers • Importance of differentiating types of fire extinguishers	

PERFORMANCE CRITERIA	ELEMENTS	COMPETENCE ASSESSMENT			TRAINING REQUIREMENTS
		PROCESS ASSESSMENT	SERVICE ASSESSMENT	UNDERPINNING KNOWLEDGE ASSESSMENT	
				Circumstantial knowledge: **Detailed knowledge about** • Safety precautions while handling fire accidents • Safe handling of tools and equipment • Firefighting rules and regulations	

LEVEL: I

MODULE: 1 **MODULE TITLE MAINTAINING SAFETY OF WORKSHOP AND SURROUNDING**

UNIT NO. 4 **UNIT TITLE:** **PERFORM FIRST AID** **DURATION: 15HRS**

PERFORMANCE CRITERIA	ELEMENTS	PROCESS ASSESSMENT	COMPETENCE ASSESSMENT		
			Service ASSESSMENT	**UNDERPINNING KNOWLEDGE ASSESSMENT**	**TRAINING REQUIREMENTS**
A person who has achieved this unit should be able to perform first aid as per requirements.	1. Perform artificial respiratory 2. Perform First aid to minor wound scalpels.	**The trainee should be able to:** • Select tools and equipment. • Identify types of injuries. • Perform artificial respiration. • Attend minor wounds. • Sterilize first aid tools. • Observe safety precautions. • Store first aid kit.	First aid offered conforms to medical requirements.	**Detailed Knowledge of:** **Method used:** The trainee should explain how to perform first aid. **Principles:** The trainee should explain principles of:- • Performing artificial respiration. • Attending minor wounds. • Providing first aid. **Theories:** The trainee should explain:- • Different types of wounds. • Different types of accidents. • Types of artificial respiration.	This unit can be achieved at a work place or Training institution. The following tools, equipment and safety gears should be available:- • First aid Kit. • Stretcher. • Light blanket. • Sterilizer. • Towel • Over-Coat. • Medical gloves. • Safety boots. • Dust Mask.

PERFORMANCE CRITERIA	ELEMENTS	PROCESS ASSESSMENT	COMPETENCE ASSESSMENT		
			Service ASSESSMENT	UNDERPINNING KNOWLEDGE ASSESSMENT	TRAINING REQUIREMENTS
				• The use of accessories in a first aid kit. • Importance of first aid. **Circumstantial knowledge** **Detailed knowledge about:** • Safety precautions to be observed while performing first aid. • Safe handling of first aid kit. • Waste disposal.	

LEVEL: I

MODULE: 2 **MODULE TITLE: PERFORM PREVENTIVE MAINTANANCE OF TOOLS, EQUIPMENT AND COMPUTER [DURATION: 80HRS]**

UNIT NO. 1 **UNIT TITLE: PERFORM PREVENTIVE MAINTENANCE OF COMPUTER TOOLS DURATION: 20HRS**

PERFORMANCE CRITERIA	ELEMENTS	PROCESS ASSESSMENT	COMPETENCE ASSESSMENT		
			PRODUCT ASSESSMENT	UNDERPINNING KNOWLEDGE ASSESSMENT	TRAINING REQUIREMENTS
A person who has achieved this unit should be able to perform maintenance of equipment as per specifications.	1. Maintaining Workshop safety Gears 2. Maintaining measuring tool 3. Maintaining Cutting and network tools	**The trainee should be able to:** • Select tools and equipment. • Interpret maintenance schedule chart. • Identify faults. • Service tools. • Observe safety precautions. • Clean tools, and workplace. • Store tools.	Maintained Tools functions as per manufacturer's specifications.	**Detailed Knowledge of:** **Method used:** The trainee should explain different ways of maintaining tools. **Principles:** The trainee should explain principles of maintaining tools. **Theories:** The trainee should explain: • Preventive maintenance. • Corrective maintenance. • Importance of maintenance schedule. • Preparation of warning tags.	This unit can be achieved at a work place or training institution. The following tools, should be available:- • Assorted power operated tools. • Assorted manually operated tools. • Computer Tool kit. • Maintenance schedule chart. • Waste bin. • Blower. • Sprit can. • Safety clear glasses. • Gloves. • Over Coat.

PERFORMANCE CRITERIA	ELEMENTS	PROCESS ASSESSMENT	COMPETENCE ASSESSMENT		
			PRODUCT ASSESSMENT	UNDERPINNING KNOWLEDGE ASSESSMENT	TRAINING REQUIREMENTS
				Circumstantial knowledge **Detailed knowledge about:** • Safety precautions to be observed while maintaining tool. • Safe handling of tools. • Waste disposal.	

LEVEL: I

MODULE: 2 **MODULE TITLE: PERFORM PREVENTIVE MAINTANANCE OF TOOLS, EQUIPMENT AND COMPUTER**

UNIT NO. 2 **UNIT TITLE: PERFORM PREVENTIVE MAINTENANCE OF COMPUTER EQUIPMENT'S** **DURATION: 20HRS**

PERFORMANCE CRITERIA	ELEMENTS	PROCESS ASSESSMENT	COMPETENCE ASSESSMENT		
			PRODUCT ASSESSMENT	UNDERPINNING KNOWLEDGE ASSESSMENT	TRAINING REQUIREMENTS
A person who has achieved this unit should be able to maintain workshop tools as per rules and regulations.	1. Maintaining Passive equipment's 2. Maintaining Static equipment 3. Maintaining Active Equipment's	**The trainee should be able to:** • Select tools, equipment and safety gears. • Categorize Equipment. • Identify Equipment's faults. • Rectify faulty Equipment. • Perform greasing. • Perform oiling. • Observe safety precautions. • Clean equipment. • Store equipment.	Maintained tools conform to manufacturer's specifications.	**Detailed Knowledge of:** **Method used:** The trainee should explain different ways of maintaining Equipment. **Principles:** The trainee should explain the principles of: • Sharpening Equipment. • Lubricating equipment. • Storing equipment. **Theories:** The trainee should explain: • Safety precautions of equipment's. • Types of maintenance. • Steps of sharpening. • Application of lubricants.	This unit can be achieved at a work place or training institution. The following tools, equipment and safety gears, should be available:- • Store room. • Tool racks. • Cabinets. • Tool boxes. • Tool shelves. • Work bench. • Service manuals. • Store ledgers. • Tool tray. • Sprit can. • Over-coat. • Gloves. • Safety clear glasses.

PERFORMANCE CRITERIA	ELEMENTS	PROCESS ASSESSMENT	COMPETENCE ASSESSMENT		
			PRODUCT ASSESSMENT	UNDERPINNING KNOWLEDGE ASSESSMENT	TRAINING REQUIREMENTS
				Circumstantial knowledge **Detailed knowledge about:** • Safety precautions to be observed while maintaining Equipments. • Waste disposal.	

LEVEL: I

MODULE: 2 **MODULE TITLE: PERFORM PREVENTIVE MAINTANANCE OF TOOLS, EQUIPMENT AND COMPUTER**

UNIT NO. 3 **UNIT TITLE: PERFORM PREVENTIVE MAINTENANCE OF COMPUTER** **DURATION: 40HRS**

PERFORMANCE CRITERIA	ELEMENTS	PROCESS ASSESSMENT	COMPETENCE ASSESSMENT		
			PRODUCT ASSESSMENT	UNDERPINNING KNOWLEDGE ASSESSMENT	TRAINING REQUIREMENTS
A person who has achieved this unit should be able to maintain workshop tools as per rules and regulations.	1. Perform General Cleaning of Computer 2. Using Computer Utility. 3. Remove Dry Joints. 4. Create Restore Point	**The trainee should be able to:** • Use service manual • Select tools and equipment for cleaning • Interpret computer Utility • Create and select Restore Point • Check functionality basic utility • Identify fault by scanning • Perform Dust cleaning. • Perform Form Cleaning. • Observe safety. • Identify Dry Joint. • Remove Dry joint. • Handle Computer .	Maintained tools conform to manufacturer's specifications.	**Detailed Knowledge of:** **Method used:** The trainee should explain different ways of maintaining tools. **Principles:** The trainee should explain the principles of: • Preventing computer From crush • Handle the Computer **Theories:** The trainee should explain: • Types of maintenance. • Function of different types of Utility tools • Importance of preventive maintenance of Computer	This unit can be achieved at a work place or training institution. The following tools, equipment and safety gears, should be available:- • Store room. • Tool racks. • Cabinets. • Tool boxes. • Tool shelves. • Work bench. • Service manuals. • Store ledgers. • Tool tray. • Vice. • Sprit can. • Over-coat. • Gloves. • Safety clear glasses.

PERFORMANCE CRITERIA	ELEMENTS	PROCESS ASSESSMENT	COMPETENCE ASSESSMENT		
			PRODUCT ASSESSMENT	UNDERPINNING KNOWLEDGE ASSESSMENT	TRAINING REQUIREMENTS
				• Importance of service manuals in preventive maintenance of Computer **Circumstantial knowledge** **Detailed knowledge about:** • Safety precautions while handling Computer • Safe way using Computer Utility • Waste disposal procedures	

LEVEL: I

MODULE: 3 **MODULE TITLE: PERFORMING ELECTRICAL JOINTS**

UNIT NO. 1 **UNIT TITLE: PERFORM COLD ELECTRICAL JOINTS** **DURATION: 12HRS**

PERFORMANCE CRITERIA	ELEMENTS	PROCESS ASSESSMENT	COMPETENCE ASSESSMENT		
			PRODUCT ASSESSMENT	UNDERPINNING KNOWLEDGE ASSESSMENT	TRAINING REQUIREMENTS
A person who has achieved this unit should be able to perform cold electrical joints as per I.E.E. regulations.	1. Make eyelet joint 2. Make crimp Joint 3. Make parallel groove joint. 4. Make bolt Joint 5. Make twist joint 6. Make Tee Joint 7. Make married joint.	**The trainee should be able to:** • Select tools, equipment and materials. • Prepare cables for joining. • Make cable joints. • Test joints. • Clean tools, equipment and workplace. • Store tools, equipment and components.	Joint made is mechanically and electrically sound.	**Detailed Knowledge of:** **Method used:** The trainee should explain how to: • Perform eyelet joint. • Perform crimp joint. • Perform clamp joint. • Perform bolt joint. **Principles:** The trainee should explain the principle of making cold electrical joints. **Theories:** The trainee should explain: • Type of joints and their application. • Materials used in making joints. • Soundings of an electrical joint.	This unit can be achieved at a work place or training institution. The following tools, safety gears and equipment should be available: • Digital and analogue multimeters. • Tool kit. • Work bench. • Safety goggles. • Safety boots. • Overall/overcoat.

PERFORMANCE CRITERIA	ELEMENTS	PROCESS ASSESSMENT	COMPETENCE ASSESSMENT		
			PRODUCT ASSESSMENT	UNDERPINNING KNOWLEDGE ASSESSMENT	TRAINING REQUIREMENTS
				Circumstantial knowledge **Detailed knowledge about:** • Safety measures involved in making joint. • Safe handling of tools and equipment.	

LEVEL: **I**

MODULE: **3** **MODULE TITLE: PERFORMING ELECTRICAL JOINTS**

UNIT NO. **2** **UNIT TITLE: PERFORM SOLDERING** **DURATION: 8HRS**

PERFORMANCE CRITERIA	**ELEMENTS**	**PROCESS ASSESSMENT**	**COMPETENCE ASSESSMENT**		
			PRODUCT ASSESSMENT	**UNDERPINNING KNOWLEDGE ASSESSMENT**	**TRAINING REQUIREMENTS**
A person who has achieved this unit should be able to perform cold electrical joints as per I.E.E. regulations.	2.1. Carry out soft soldering 2.2. Carry out hard soldering.	**The trainee should be able to:** • Interpret the diagrams. • Identify tools, safety gears, equipment and materials required. • Prepare parts to be soldered. • Solder joints. • Remove surplus solder using wire brush. • Clean work area. • Store tools, safety gears, equipment and materials.	Soldered joint conforms to technical specifications.	**Detailed Knowledge of:** **Method used:** The trainee should explain how to: • Solder by using pot and ladle. • Solder by using soldering iron. • Solder by using soldering gun. **Principles:** The trainee should explain the principle of: • Soft soldering. • Hard soldering. **Theories:** The trainee should explain: • Different types of soldering. • Type of soldering materials. • Application of soldering.	This unit can be achieved at work place or training institution. The following tools, safety gears and equipment should be available: • Tool kit. • Blow lamp. • Soldering stand. • Analogue and digital multimeters. • Overall/overcoat. • Safety goggles. • Safety boots.

PERFORMANCE CRITERIA	ELEMENTS	PROCESS ASSESSMENT	COMPETENCE ASSESSMENT		
			PRODUCT ASSESSMENT	UNDERPINNING KNOWLEDGE ASSESSMENT	TRAINING REQUIREMENTS
				Circumstantial knowledge **Detailed knowledge about:** • Safety precautions during soldering. • Safe handling of tools and equipment.	

LEVEL: I

MODULE NUMBER: 4 **MODULE TITLE: BUILDING SIMPLE ELECTRIC CIRCUITS [DURATION: 20HRS]**

UNIT NO: 1 **UNIT TITLE: CONSTRUCT RESISTIVE CIRCUITS** **DURATION: 4 HOURS.**

PERFORMANCE CRITERIA	**ELEMENTS**	**COMPETENCE ASSESSMENT**			**TRAINING REQUIREMENTS**
		PROCESS ASSESSMENT	**PRODUCT ASSESSMENT**	**UNDERPINNING KNOWLEDGE ASSESSMENT**	
A person who has achieved this unit should be able to construct resistive circuit as per given electronic drawing and configuration.	1. Build single resistor circuit. 2. Build series resistive circuit. 3. Build parallel resistive circuit. 4. Build combination circuit.	**The trainee should be able to:** • Interpret circuit diagram. • Select tools, equipment and material. • Select components. • Construct single resistor circuit. • Construct series resistive circuit. • Construct combination circuit. • Solder constructed circuits. • Test built circuits.	Constructed circuits function as per technical specifications.	**Knowledge evidence:** **Detailed knowledge of:** **Method used:** The trainee should explain how to: • Measure values of resistance. • Read resistor values by colour codes. • Connect resistors in series and parallel. **Principles:** The trainee should explain principle of: • Constructing resistive circuits. • Performing measurements in resistor circuits.	This unit can be achieved at a work place or training institution. The following tools, equipment and safety gears should be available: • Work bench. • Electronics board. • Tool kit. • Measuring tape. • Analogue and digital multimeters. • Overall. • Gloves • Safety goggles. • Safety boots.

PERFORMANCE CRITERIA	ELEMENTS	COMPETENCE ASSESSMENT			TRAINING REQUIREMENTS
		PROCESS ASSESSMENT	**PRODUCT ASSESSMENT**	**UNDERPINNING KNOWLEDGE ASSESSMENT**	
		• Clean tools, equipment and work place. • Store tools, equipment and materials.		**Theories:** The trainee should explain: • Concept of electricity • Characteristics of different resistive circuits. • Application of Ohm's law. • Electrical symbols. **Circumstantial knowledge:** **Detailed knowledge about:** • Safe handling of work tools and equipment. • Electrical hazards. • First aid.	

LEVEL: I

MODULE NUMBER:4 **MODULE TITLE: BUILDING SIMPLE ELECTRIC CIRCUITS**

UNIT NO: 2 **UNIT TITLE: CONSTRUCT CAPACITIVE CIRCUITS** **DURATION: 4 HOURS.**

PERFORMANCE CRITERIA	ELEMENTS	COMPETENCE ASSESSMENT			TRAINING REQUIREMENTS
		PROCESS ASSESSMENT	PRODUCT ASSESSMENT	UNDERPINNING KNOWLEDGE ASSESSMENT	
A person who has achieved this unit should be able to construct capacitive circuit as per given circuit diagram.	1. Build single capacitor circuit. 2. Build series inductive circuit. 3. Build parallel capacitive circuits. 4. Build combination circuit.	**The trainee should be able to:** • Interpret circuit diagram of capacitive circuit. • Select tools, equipment and material required. • Prepare capacitors and cables for termination. • Construct circuit on board. • Solder built circuits. • Test circuit. • Record test results.	Constructed capacitive circuit functions as per technical specifications.	**Knowledge evidence:** **Detailed knowledge of:** **Method used:** The trainee should explain how to: • Connect capacitors in series and parallel. • Solder capacitors. • Measure capacitance values. • Identify different capacitors. • Calculate value of capacitive reactance.	This unit can be achieved at a work place or training institution. The following tools, equipment and safety gears should be available: • Electronics board. • Tool kit. • Analogue and digital multimeters. • Measuring tape. • Overall. • Safety goggles. • Work bench. • Safety boots.

PERFORMANCE CRITERIA	ELEMENTS	COMPETENCE ASSESSMENT			TRAINING REQUIREMENTS
		PROCESS ASSESSMENT	**PRODUCT ASSESSMENT**	**UNDERPINNING KNOWLEDGE ASSESSMENT**	
		• Clean tools, equipment and work place. • Store tools, equipment and materials.		**Principles:** The trainee should explain the principles of: • Reading capacitor colour code. • Constructing capacitive circuits. • Measuring capacitance. **Theories:** The trainee should explain: • Characteristics of capacitors. • Application of different types of capacitors. • Electrical symbols.	

PERFORMANCE CRITERIA	ELEMENTS	COMPETENCE ASSESSMENT			TRAINING REQUIREMENTS
		PROCESS ASSESSMENT	PRODUCT ASSESSMENT	UNDERPINNING KNOWLEDGE ASSESSMENT	
				Circumstantial knowledge: **Detailed knowledge about:** • Safety precautions in handling capacitors. • Safe handling of work tools and equipment.	

LEVEL: I

MODULE NUMBER: 4 **MODULE TITLE: BUILDING SIMPLE ELECTRIC CIRCUITS**

UNIT NO: 3 **UNIT TITLE: CONSTRUCT INDUCTIVE CIRCUITS** **DURATION: 4 HOURS.**

PERFORMANCE CRITERIA	ELEMENTS	COMPETENCE ASSESSMENT			TRAINING REQUIREMENTS
		PROCESS ASSESSMENT	PRODUCT ASSESSMENT	UNDERPINNING KNOWLEDGE ASSESSMENT	
A person who has achieved this unit should be able to construct inductive circuit as per electronics common rules and configuration.	1. Build single inductor circuit. 2. Build series inductive circuit. 3. Build parallel inductive circuit. 4. Build combination circuit.	**The trainee should be able to:** • Interpret diagram of inductive circuit. • Select tools, equipment and material. • Prepare inductors and cables for termination. • Build inductor circuits. • Solder built circuits. • Test built circuit. • Record test results. • Clean tools, equipment and work place.	Constructed inductive circuit conforms to electrical specifications.	**Knowledge evidence:** **Detailed knowledge of:** **Method used:** The trainee should explain how to: • Identify an inductor. • Connect inductors in series and parallel. • Measure values of inductance. • Calculate value of inductive reactance. **Principles:** The trainee should explain the principles of: • Constructing inductive circuits. • Measuring inductance.	This unit can be achieved at a work place or training institution. The following tools, equipment and safety gears should be available: • Electronics board. • Tool kit. • Analogue and digital multimeters. • Measuring tape. • Overall. • Safety goggles. • Work bench. • Safety boots.

PERFORMANCE CRITERIA	ELEMENTS	COMPETENCE ASSESSMENT			TRAINING REQUIREMENTS
		PROCESS ASSESSMENT	**PRODUCT ASSESSMENT**	**UNDERPINNING KNOWLEDGE ASSESSMENT**	
		• Store tools and equipment.		**Theories:** The trainee should explain: • Characteristics of inductors. • Application of different types of inductive circuits. • Electrical symbols. **Circumstantial knowledge:** **Detailed knowledge about:** • Safety precautions in inductive loads. • Safe handling of tools, equipment and inductors.	

LEVEL: I

MODULE NUMBER: 4 **MODULE TITLE: BUILDING SIMPLE ELECTRICAL CIRCUITS**

UNIT NO: 6 **UNIT TITLE CONSTRUCTING RLC CIRCUIT** **DURATION: 6 HOURS.**

<table>
<tr><th rowspan="2">PERFORMANCE CRITERIA</th><th rowspan="2">ELEMENTS</th><th colspan="3">COMPETENCE ASSESSMENT</th><th rowspan="2">TRAINING REQUIREMENTS</th></tr>
<tr><th>PROCESS ASSESSMENT</th><th>PRODUCT ASSESSMENT</th><th>UNDERPINNING KNOWLEDGE ASSESSMENT</th></tr>
<tr><td>A person who has achieved this unit should be able to construct RLC circuit as per given circuit diagram.</td><td>1. Build resistance and capacitance circuit.
2. Build resistance and inductance circuit.
3. Build resistance, capacitance and inductance circuit.</td><td>The trainee should be able to:
• Interpret diagram of RLC circuit.
• Select tools, equipment and materials.
• Prepare inductors, capacitors and resistors.
• Construct circuit on board using inductor, capacitor and resistor.
• Solder built circuit.
• Test circuit.
• Record test results.</td><td>Constructed RLC circuit functions as per technical specifications.</td><td>Knowledge evidence:
Detailed knowledge of:
Method used: The trainee should explain how to:
• Construct RLC circuit.
• Measure value of resistor, capacitor and inductor.
• Calculate impedance of RLC circuit.
Principles: The trainee should explain the principles of:
• Constructing RLC circuit.
• Measuring value of impedance.
Theories: The trainee should explain:
• RLC circuit and its behaviour.
• Application of RLC circuit.</td><td>This unit can be achieved at a work place or training institution.
The following tools, equipment and safety gears should be available:
• Electronics board.
• Tool kit.
• Analogue and digital multimeters.
• Measuring tape.
• Overall.
• Safety goggles.
• Work bench.
• Safety boots.</td></tr>
</table>

PERFORMANCE CRITERIA	ELEMENTS	COMPETENCE ASSESSMENT			TRAINING REQUIREMENTS
		PROCESS ASSESSMENT	**PRODUCT ASSESSMENT**	**UNDERPINNING KNOWLEDGE ASSESSMENT**	
		• Clean tools, equipments and work place. • Store tools, equipment and materials.		**Circumstantial knowledge:** **Detailed knowledge about:** • Safety precautions when soldering. • Safe handling of tools and RLC components.	

LEVEL: I

MODULE NUMBER: 4 **MODULE TITLE: BUILDING SIMPLE ELECTRIC CIRCUITS**

UNIT NO: 5 **UNIT TITLE: MEASURE ELECTRIC QUANTITIES** **DURATION: 2 HOURS.**

PERFORMANCE CRITERIA	ELEMENTS	COMPETENCE ASSESSMENT			TRAINING REQUIREMENTS
		PROCESS ASSESSMENT	**PRODUCT ASSESSMENT**	**UNDERPINNING KNOWLEDGE ASSESSMENT**	
A person who has achieved this unit should be able to measure electrical quantities as per set standards.	1. Measure voltage in the circuit. 2. Measure current in the circuit. 3. Measure resistance in the circuit. 4. Measure power in the circuit.	**The trainee should be able to:** • Determine component values. • Connect simple electric circuits. • Perform soldering. • Measure electric quantities. • Record measured values. • Clean tools, equipment and work place. • Store tools and equipment.	Recorded values conform to standard values.	**Knowledge evidence:** **Detailed knowledge of:** **Method used:** The trainee should explain how to: • Measure components values. • Measure values of voltage and current in circuit. **Principles:** The trainee should explain principle of: • Reading colour codes. • Performing soldering. • Measuring electric quantities.	This unit can be achieved at a work place or training institution. The following tools, equipment and safety gears should be available: • Electronics board. • Analog and digital Multimeters. • Tool kit. • Work bench. • Work bench light. • Power supply. • Safety boots. • Safety gloves. • Overall.

PERFORMANCE CRITERIA	ELEMENTS	COMPETENCE ASSESSMENT			TRAINING REQUIREMENTS
		PROCESS ASSESSMENT	PRODUCT ASSESSMENT	UNDERPINNING KNOWLEDGE ASSESSMENT	
				Theories: The trainee should explain: • Ohm's law. • Different component ratings. • Types of electric circuit connections. • Verification of electric rules and laws. • Importance of component ratings. • Types and uses of measuring equipment. • Application of colour codes.	

PERFORMANCE CRITERIA	ELEMENTS	COMPETENCE ASSESSMENT			TRAINING REQUIREMENTS
		PROCESS ASSESSMENT	PRODUCT ASSESSMENT	UNDERPINNING KNOWLEDGE ASSESSMENT	
				Circumstantial knowledge: **Detailed knowledge about:** • Safe handling of working tools. • Safe handling of measuring instruments.	

LEVEL: I

MODULE NUMBER: 5 **MODULE TITLE: BUILDING SIMPLE ELECTRONIC CIRCUITS** **[DURATION 80HRS]**

UNIT NO: 1 **UNIT TITLE: DETERMINE CHARACTERISTICS OF ACTIVE ELECTRONIC DEVICES** **DURATION:14 HOURS.**

PERFORMANCE CRITERIA	**ELEMENTS**	**COMPETENCE ASSESSMENT**			**TRAINING REQUIREMENTS**
		PROCESS ASSESSMENT	**PRODUCT ASSESSMENT**	**UNDERPINNING KNOWLEDGE ASSESSMENT**	
A person who has achieved this unit should be able to determine characteristics of electronic components as per technical specifications.	1. Test characteristics of diode. 2. Test characteristics of transistors. 3. Test characteristics of thyristors. 4. Test characteristics of opto-electronic devices. 5. Test characteristics of integrated circuits.	**The trainee should be able to:** • Select tools and equipment. • Select electronic components. • Construct circuit for component testing. • Test electronic component. • Record test results. • Interpret standard test results. • Observe safety regulations. • Clean tools, equipment and workplace.	Tested components bear characteristics that conform to specifications as given in component data books.	**Knowledge evidence:** **Detailed knowledge of:** **Method used:** The trainee should explain how to test components. **Principles:** The trainee should explain the principles of: • Operating test equipment and measuring instruments. • Testing components. **Theories:** The trainee should explain: • Types of active electronic components. • The difference between passive and active electronic components.	This unit can be achieved at a workplace or training institution. The following tools, equipment and safety gears should be available: • Digital and analogue multimeters. • Oscilloscope. • Curve tracer. • Tool kit. • Work bench. • Gloves. • Overcoat. • Overall. • Boots.

PERFORMANCE CRITERIA	ELEMENTS	COMPETENCE ASSESSMENT			TRAINING REQUIREMENTS
		PROCESS ASSESSMENT	**PRODUCT ASSESSMENT**	**UNDERPINNING KNOWLEDGE ASSESSMENT**	
		• Store tools, equipment and components.		• Characteristics of active electronic components. • Variation of component performance with temperature. **Circumstantial knowledge:** **Detailed knowledge about:** • Safety precautions in electronic work. • Safe handling of tools, test equipment and measuring instruments. • Safe handling of electronic components.	

LEVEL: I

MODULE NUMBER: 5 **MODULE TITLE: BUILDING SIMPLE ELECTRONIC CIRCUITS**

UNIT NO: 2 **UNIT TITLE: BUILD RECTIFIER CIRCUITS** **DURATION: 26 HOURS.**

<table>
<tr><th rowspan="2">PERFORMANCE CRITERIA</th><th rowspan="2">ELEMENTS</th><th colspan="3">COMPETENCE ASSESSMENT</th><th rowspan="2">TRAINING REQUIREMENTS</th></tr>
<tr><th>PROCESS ASSESSMENT</th><th>PRODUCT ASSESSMENT</th><th>UNDERPINNING KNOWLEDGE ASSESSMENT</th></tr>
<tr><td>A person who has achieved this unit should be able to build rectifier circuit to produce D.C. voltage as per technical specifications.</td><td>1. Build half-wave rectifier.
2. Build full-wave rectifier (centre tapped).
3. Build bridge rectifier.
4. Build a smoothing circuit.</td><td>The trainee should be able to:
• Interpret circuit diagrams.
• Select tools, equipment and components.
• Construct rectifier circuit on board.
• Perform soldering.
• Test built rectifier circuits.
• Record test results.
• Observe safety precautions.
• Clean tools, equipment and work place.</td><td>Built rectifier circuit produces expected voltage.</td><td>Knowledge evidence:
Detailed knowledge of:
Method used: The trainee should explain how to construct a D.C. power supply.
Principles: The trainee should explain the principles of:
• Constructing D.C. power supply.
• Performing soldering.
• Rectifying A.C. voltage.</td><td>This unit can be achieved at a workplace or training institution.
The following tools, equipment and safety gears should be available:
• Digital/analogue multimeter.
• Oscilloscope.
• Tool kit.
• Soldering iron.
• Work bench.
• Work bench light.
• Magnifying glass.
• Gloves.
• Overcoat/overall.
• Boots.</td></tr>
</table>

PERFORMANCE CRITERIA	ELEMENTS	COMPETENCE ASSESSMENT			TRAINING REQUIREMENTS
		PROCESS ASSESSMENT	**PRODUCT ASSESSMENT**	**UNDERPINNING KNOWLEDGE ASSESSMENT**	
		• Store tools and equipment.		**Theories:** The trainee should explain: • Characteristics of dc power supply components. • Application of rectifier circuits. **Circumstantial knowledge:** **Detailed knowledge about:** • Safety precautions in building D.C. power supplies. • Safe handling of work tools and equipment.	

LEVEL: I

MODULE NUMBER: 5 **MODULE TITLE: BUILDING SIMPLE ELECTRONIC CIRCUITS**

UNIT NO: 3 **UNIT TITLE: SERVICE BATTERIES** **DURATION: 10 HOURS.**

<table>
<tr><th rowspan="2">PERFORMANCE CRITERIA</th><th rowspan="2">ELEMENTS</th><th colspan="3">COMPETENCE ASSESSMENT</th><th rowspan="2">TRAINING REQUIREMENTS</th></tr>
<tr><th>PROCESS ASSESSMENT</th><th>PRODUCT ASSESSMENT</th><th>UNDERPINNING KNOWLEDGE ASSESSMENT</th></tr>
<tr><td>A person who has achieved this unit should be able to service battery according to manufacturer's guidelines.</td><td>1. Charge battery.
2. Test battery.</td><td>The trainee should be able to:
• Select tools and equipment.
• Select power supply for charging battery.
• Charge battery.
• Measure battery output voltage.
• Observe safety precautions.
• Clean tools, equipment and work place.
• Store tools and equipment.</td><td>Charged battery produces rated voltage and current.</td><td>Knowledge evidence:
Detailed knowledge of:
Method used: The trainee should explain how to charge battery.
Principles: The trainee should explain the principles of:
• Maintaining stable output of a D.C. power supply unit.
• Measuring output voltage of a D.C. power supply unit.
• Measuring terminal voltage of a battery.</td><td>This unit can be achieved at a workplace or training institution. The following tools, equipment and safety gears should be available:
• D.C. power supply unit.
• Digital and analogue multimeters.
• Oscilloscope.
• Tool kit.
• Work bench.
• Battery.
• Hydrometer.
• Gloves.
• Overcoat.
• Overall.
• Boots.</td></tr>
</table>

PERFORMANCE CRITERIA	ELEMENTS	COMPETENCE ASSESSMENT			TRAINING REQUIREMENTS
		PROCESS ASSESSMENT	PRODUCT ASSESSMENT	UNDERPINNING KNOWLEDGE ASSESSMENT	
				Theories: The trainee should explain: • Parameters of D.C. power supply circuits. • Charge and discharge characteristics of battery. **Circumstantial knowledge:** **Detailed knowledge about:** • Safety precautions on handling batteries. • Safe handling of tools and equipment.	

LEVEL: I

MODULE NUMBER: 5 **MODULE TITLE: BUILDING SIMPLE ELECTRONIC CIRCUITS**

UNIT NO: 4 **UNIT TITLE: PERFORM MEASUREMENTS ON SIMPLE LOW/HIGH FREQUENCY CIRCUITS**

DURATION: 10 HOURS.

PERFORMANCE CRITERIA	**ELEMENTS**	**COMPETENCE ASSESSMENT**			**TRAINING REQUIREMENTS**
		PROCESS ASSESSMENT	**SERVICE ASSESSMENT**	**UNDERPINNING KNOWLEDGE ASSESSMENT**	
A person who has achieved this unit should be able to perform measurements on simple low/high frequency circuits as per technical specifications.	1. Test low frequency circuit. 2. Test high frequency circuit.	**The trainee should be able to:** • Select tools and equipment. • Select active electronic components. • Construct low or high frequency circuit. • Use test equipment. • Measure the circuit parameter. • Record measured results.	Measured circuit parameters conform to technical specifications.	**Knowledge evidence:** **Detailed knowledge of:** **Method used:** The trainee should explain how to test low or high frequency circuit. **Principles:** The trainee should explain the principles of: • Operating test equipment and measuring instruments. • Testing electronic circuits.	This unit can be achieved at a workplace or training institution. The following tools, equipment and safety gears should be available: • Digital and analogue multimeters. • Oscilloscope. • Frequency meter. • Curve tracer. • Power supply unit. • Tool kit. • Work bench. • Gloves. • Overcoat or overall. • Boots.

PERFORMANCE CRITERIA	ELEMENTS	COMPETENCE ASSESSMENT			TRAINING REQUIREMENTS
		PROCESS ASSESSMENT	**SERVICE ASSESSMENT**	**UNDERPINNING KNOWLEDGE ASSESSMENT**	
		• Observe safety precautions. • Clean tools, equipment and work place. • Store tools and equipment.		**Theories:** The trainee should explain: • Difference between low and high frequency circuits. • Characteristics of low frequency circuit. • Characteristics of high frequency circuit. **Circumstantial knowledge:** **Detailed knowledge about:** • Safety precautions in electronic work. • Safe handling of test equipment and measuring instruments.	

LEVEL: I

MODULE NUMBER: 5 **MODULE TITLE: BUILDING SIMPLE ELECTRONIC CIRCUITS**

UNIT NO: 5 **UNIT TITLE: TROUBLESHOOT ANALOGUE ELECTRONIC CIRCUITS** **DURATION: 20 HOURS.**

PERFORMANCE CRITERIA	**ELEMENTS**	**COMPETENCE ASSESSMENT**			**TRAINING REQUIREMENTS**
		PROCESS ASSESSMENT	**SERVICE ASSESSMENT**	**UNDERPINNING KNOWLEDGE ASSESSMENT**	
A person who has achieved this unit should be able to troubleshoot analogue electronic circuits as per technical guidelines.	1. Troubleshoot stabilized power supply. 2. Troubleshoot single stage transistor amplifier. 3. Troubleshoot multi-stage transistor and amplifier.	**The trainee should be able to:** • Select tools and equipment. • Troubleshoot analogue electronic circuit. • Identify defective components. • Replace defective components. • Observe safety precautions. • Test circuit • Clean tools, equipment and workplace. • Store tools, equipment and components.	Analogue electronic circuit serviced as per technical specifications.	**Knowledge evidence:** **Detailed knowledge of:** **Method used:** The trainee should explain how to troubleshoot analogue circuits. **Principles:** The trainee should explain the principles of: • Constructing analogue electronic circuits. • Troubleshooting analogue electronic circuits. • Operating test equipment and measuring instruments.	This unit can be achieved at a workplace or training institution. The following tools, equipment and safety gears should be available: • Digital and analogue multimeters. • Oscilloscope. • D.C. Power supply unit. • Signal generator. • Tool kit. • Work bench. • Gloves. • Overcoat or overall. • Safety boots.

PERFORMANCE CRITERIA	ELEMENTS	COMPETENCE ASSESSMENT			TRAINING REQUIREMENTS
		PROCESS ASSESSMENT	SERVICE ASSESSMENT	UNDERPINNING KNOWLEDGE ASSESSMENT	
				Theories: The trainee should explain: • Performance of electronic components. • Types of electronic circuits. • Parameters of analogue electronic circuits. • Single and multi-stage transistor amplifiers. **Circumstantial knowledge:** **Detailed knowledge about:** • Safety precautions in electronic work. • Safe handling of tools, test equipment and measuring instruments. • Safe handling of electronic components.	

LEVEL: **I**

MODULE: **6** **MODULE TITLE: MAINTAINING COMPUTER AND ITS PERIPHERALS.** **DURATION: 190 HOURS**

UNIT NO. **1** **UNIT TITLE: INSTALLATION OF COMPUTER AND ITS PERIPHERALS.** **DURATION: 110 HOURS**

<table>
<tr><th rowspan="2">PERFORMANCE CRITERIA</th><th rowspan="2">ELEMENTS</th><th rowspan="2">PROCESS ASSESSMENT</th><th colspan="3">COMPETENCE ASSESSMENT</th></tr>
<tr><th>Service ASSESSMENT</th><th>UNDERPINNING KNOWLEDGE ASSESSMENT</th><th>TRAINING REQUIREMENTS</th></tr>
<tr>
<td>A person who has achieved this unit should be able to arrange and connect microcomputer system and its peripherals according to the installation manual.</td>
<td>1. Install of computer and its peripherals.
2. Troubleshooting of computer peripherals.</td>
<td>The trainee should be able to:
• Identify parts of microcomputer set and its peripherals
• Arrange the parts of microcomputer set and its peripherals
• Identify power and data cables
• Identify the connection ports
• Connect the parts of microcomputer set and its peripherals together
• Connect the computer system to power source
• Test the connected system</td>
<td>Connected computer system organized and working as per installation manuals.</td>
<td>Knowledge evidence:
Detailed Knowledge of:
Method used:
The trainee should explain:
• Different connections in different types of microcomputer system.
• The process of connecting microcomputer system.
Principles:
The trainee should explain the principles involved in microcomputer system connections.
Theories:
The trainee should explain:
• The trainee should explain the functions of peripherals and computer system.</td>
<td>This can be achieved at work place or at institution.
The following equipment should be available:
• UPS
• Computer set
• Power and data cables
• Printer
• Scanner</td>
</tr>
</table>

PERFORMANCE CRITERIA	ELEMENTS	PROCESS ASSESSMENT	COMPETENCE ASSESSMENT		
			Service ASSESSMENT	UNDERPINNING KNOWLEDGE ASSESSMENT	TRAINING REQUIREMENTS
				• The trainee should describe different types of computers. • Different types of peripherals. **Circumstantial Knowledge:** **Detailed knowledge about:** • Confidentiality, Trustworthy and security. • Safety precautions for the machine and human.	

LEVEL: 1

MODULE: 6 **MODULE TITLE: MAINTAINING COMPUTER AND ITS PERIPHERALS**

UNIT NO. 2 **UNIT TITLE: COMPUTER ASSEMBLING** **DURATION: 80 HOURS**

PERFORMANCE CRITERIA	ELEMENTS	PROCESS ASSESSMENT	COMPETENCE ASSESSMENT		
			SERVICE ASSESSMENT	UNDERPINNING KNOWLEDGE ASSESSMENT	TRAINING REQUIREMENTS
A person who has achieved this unit should be able to assemble computer system according to installation manuals.	1. Connecting computer components. 2. Upgrading computer components. 3. Troubleshooting of computer components	**The trainee should be able to:** • Discharge the body from Static voltage • Prepare computer hardware • Set jumpers/switches • Identify connection points • Connect the hardware together • Connect the computer system to power source • Test assembled computer	Computer system assembled and working as per requirements.	**Knowledge evidence:** **Detailed Knowledge of:** **Method used:** The trainee should explain the procedures for assembling computer system. **Principles:** The trainee should explain the principles involved in computer assembling. **Theories:** The trainee should explain: • The functions of different parts of a computer. • The importance of assembling computer system.	This unit can be achieved at work place or at training institution. The following tools and equipment should be available: • UPS • Computer parts • Computer tool kit

PERFORMANCE CRITERIA	ELEMENTS	PROCESS ASSESSMENT	COMPETENCE ASSESSMENT		
			SERVICE ASSESSMENT	**UNDERPINNING KNOWLEDGE ASSESSMENT**	**TRAINING REQUIREMENTS**
				Circumstantial Knowledge: **Detailed knowledge about:** • Confidentiality, Trustworthy and security • Safety precautions for the machine and human.	

LEVEL: 1

MODULE: 7 **MODULE TITLE: MAINTAINING OPERATING SYSTEM [DURATION: 190 HOURS]**

UNIT NO. 1 **UNIT TITLE: INSTALLATION OF OPERATING SYSTEM DURATION: 90 HOURS**

PERFORMANCE CRITERIA	ELEMENTS	PROCESS ASSESSMENT	COMPETENCE ASSESSMENT		
			SERVICE ASSESSMENT	UNDERPINNING KNOWLEDGE ASSESSMENT	TRAINING REQUIREMENTS
A person who has achieved this unit should be able to install and maintain the operating system software in the computer according to installation manuals.	1. Install the operating system 2. Updating of operating system 3. Troubleshooting of operating system	**The trainee should be able to:** • Check hardware specifications. • Identify operating system to be installed • Start the computer with bootable disk. • Partition the hard disk • Format the hard disk • Install the operating system. • Upgrade operating system. • Update operating system. • Recover operating system.	The operating system software installed and working as per default components of operating system.	**Knowledge evidence:** **Detailed Knowledge of:** **Method used:** The trainee should explain the different methods of installing and maintaining operating system. **Principles:** The trainee should explain the principles involved in software installation. **Theories:** The trainee should explain: • The functions of operating system software. • Different operating systems	This unit can be achieved at work place or at training institution. The following tools and equipment should be available: • Computer set • External storage device • Operating System software • Device drivers

<table>
<tr><th rowspan="2">PERFORMANCE CRITERIA</th><th rowspan="2">ELEMENTS</th><th rowspan="2">PROCESS ASSESSMENT</th><th colspan="3">COMPETENCE ASSESSMENT</th></tr>
<tr><th>SERVICE ASSESSMENT</th><th>UNDERPINNING KNOWLEDGE ASSESSMENT</th><th>TRAINING REQUIREMENTS</th></tr>
<tr><td></td><td></td><td>Configure device drivers.
Test the computer system</td><td></td><td>Circumstantial Knowledge:
Detailed knowledge about:
Confidentiality, Trustworthy and security
Safety precautions for the machine and human.</td><td></td></tr>
</table>

LEVEL: 1

MODULE: 7 **MODULE TITLE: MAINTAINING OPERATING SYSTEM**

UNIT NO. 2 **UNIT TITLE: INSTALLATION AND MAINTAINING APPLICATION PROGRAM** **DURATION: 50 HOURS**

PERFORMANCE CRITERIA	ELEMENTS	PROCESS ASSESSMENT	COMPETENCE ASSESSMENT		
			SERVICE ASSESSMENT	UNDERPINNING KNOWLEDGE ASSESSMENT	TRAINING REQUIREMENTS
A person who has achieved this unit should be able to install and maintain different types of application software as per installation manuals.	1. Install application program 2. Updating Application Program 3. Troubleshooting of application program	**The trainee should be able to:** • Switch ON the computer • Run setup • Follow installation steps • Upgrade software • Update software • Test the software	The software installed and function as per requirements.	**Knowledge evidence:** **Detailed Knowledge of:** **Method used:** The trainee should explain the different methods of installing and maintaining application software. **Principles:** • Installing and maintaining application software. • Installing different application software. **Theories:** The trainee should explain: • The importance of application software installation. • Step used to install different application software.	This unit can be achieved at work place or at training institution. The following tools and equipment should be available: • Computer set • Application software • External storage devices

PERFORMANCE CRITERIA	ELEMENTS	PROCESS ASSESSMENT	COMPETENCE ASSESSMENT		
			SERVICE ASSESSMENT	UNDERPINNING KNOWLEDGE ASSESSMENT	TRAINING REQUIREMENTS
				Circumstantial Knowledge: **Detailed knowledge about:** • Confidentiality, Trustworthy and security • Software licensing.	

LEVEL: 1

MODULE: 7 **MODULE TITLE: MAINTAINING OPERATING SYSTEM**

UNIT NO. 3 **UNIT TITLE: INSTALLATION OF DRIVERS** **DURATION: 50 HOURS**

PERFORMANCE CRITERIA	ELEMENTS	PROCESS ASSESSMENT	COMPETENCE ASSESSMENT		
			SERVICE ASSESSMENT	**UNDERPINNING KNOWLEDGE ASSESSMENT**	**TRAINING REQUIREMENTS**
A person who has achieved this unit should be able to upgrade computer hardware according to manufacturers' standards.	1. Install the drivers. 2. Updating of drivers. 3. Troubleshooting drivers.	**The trainee should be able to:** • Identify hardware to be upgraded • Determine hardware specifications • Check the compatibility of hardware • Install the upgrading hardware • Test the computer system	The computer hardware upgraded as per manufacturers' standard to increase performance that suits specific requirements.	**Knowledge evidence:** **Detailed Knowledge of:** **Method used:** The trainee should explain the different methods of upgrading the computer hardware. **Principles:** The trainee should explain the principals involved in upgrading different computer hardware. **Theories:** The trainee should explain: • The strengths and weaknesses of different computer hardware • The functions of different parts of a computer, • The importance of upgrading computer	This unit can be achieved at work place or at training institution. The following tools and equipment should be available: • Computer set • Computer hardware tool kit • Upgrading parts

PERFORMANCE CRITERIA	ELEMENTS	PROCESS ASSESSMENT	COMPETENCE ASSESSMENT		
			SERVICE ASSESSMENT	**UNDERPINNING KNOWLEDGE ASSESSMENT**	**TRAINING REQUIREMENTS**
				hardware. **Circumstantial Knowledge:** **Detailed knowledge about:** • Confidentiality, Trustworthy and security • Safety precautions in handling computer system.	

LEVEL: 1

MODULE: 8 **MODULE TITLE: MANAGING WORD PROCESSOR AND SPREADSHEET APPLICATIONS** **DURATION: 100 HOURS**

UNIT NO.1 **UNIT TITLE: PROCESSING WORD** **DURATION: 50 HOURS**

<table>
<tr><th rowspan="2">PERFORMANCE CRITERIA</th><th rowspan="2">ELEMENTS</th><th rowspan="2">PROCESS ASSESSMENT</th><th colspan="3">COMPETENCE ASSESSMENT</th></tr>
<tr><th>SERVICE ASSESSMENT</th><th>UNDERPINNING KNOWLEDGE ASSESSMENT</th><th>TRAINING REQUIREMENTS</th></tr>
<tr>
<td>A person who has achieved this unit should be able to create different documentations according to the software procedures.</td>
<td>1. Creating documents.
2. Editing and formatting document.
3. Inserting objects and images.
4. Merging documents.
5. Link Topics/Documents
6. Working with tables.</td>
<td>The trainee should be able to:
• Open new or existing word document
• Type the document
• Edit the document
• Scan image.
• Insert objects/image.
• Format the document.
• Link other topic in the existing documents.
• Save the document.
• Preview and Print the Document.</td>
<td>The document created and formatted as per secretarial standards.</td>
<td>Knowledge evidence:
Detailed Knowledge of:
Method used:
• Different methods of processing word document.
• The process of image scanning
Principles:
• Processing word document.
• Scanning image
Theories:
The trainee should explain:
• The functions and importance of word processor.
• Different types of word processors
• Different types of storage devices</td>
<td>This unit can be achieved at work place or at training institution. The following tools and equipment should be available:
• Computer set
• Word processing program
• Printer
• External storage devices
• Scanner</td>
</tr>
</table>

PERFORMANCE CRITERIA	ELEMENTS	PROCESS ASSESSMENT	COMPETENCE ASSESSMENT		
			SERVICE ASSESSMENT	UNDERPINNING KNOWLEDGE ASSESSMENT	TRAINING REQUIREMENTS
				• Different ways of opening, copying and saving documents **Circumstantial Knowledge:** **Detailed knowledge about:** • Confidentiality, trustworthy and security. • Safety precautions in handling computer system.	

LEVEL: 1

MODULE: 8 **MODULE TITLE: MANAGING WORD PROCESSOR AND SPREADSHEET APPLICATIONS**

UNIT NO. 2 **UNIT TITLE: SPREADSHEET PACKAGE** **DURATION: 50 HOURS**

<table>
<tr><th rowspan="2">PERFORMANCE CRITERIA</th><th rowspan="2">ELEMENTS</th><th rowspan="2">PROCESS ASSESSMENT</th><th colspan="3">COMPETENCE ASSESSMENT</th></tr>
<tr><th>SERVICE ASSESSMENT</th><th>UNDERPINNING KNOWLEDGE ASSESSMENT</th><th>TRAINING REQUIREMENTS</th></tr>
<tr><td>A person who has achieved this unit should be able to manipulate data and create charts using spreadsheet according to the software procedures.</td><td>1. Creating spread sheet.
2. Formatting a worksheet.
3. Performing calculation.
4. Malt sheet Calculations
5. Creating Charts.
6. Perform application programming</td><td>The trainee should be able to:
• Switch ON the computer
• Open the spreadsheet program
• Enter data in the spreadsheet
• Edit data in the spreadsheet
• Format the spreadsheet
• Manipulate data
• Sheet connection/link and calculations
• Create charts
• Use object properties and events.
• Save the spreadsheet
• Preview and Print the spreadsheet</td><td>• The data in spreadsheet manipulated as per mathematical formula.
• Charts created as per available data and requirements.</td><td>Knowledge evidence:
Detailed Knowledge of:
Method used:
• Manipulating data using spreadsheet program.
• Creating different charts using spreadsheet program
Principles:
• Manipulating data using spreadsheet.
• Creating charts using spreadsheet.
Theories:
The trainee should explain:
• Function and importance of manipulating data using spreadsheet program.
• Different types of spreadsheet programs</td><td>This unit can be achieved at work place or at training institution.
The following tools and equipment should be available:
• Computer set
• Spreadsheet program
• External storage devices
• Printer</td></tr>
</table>

PERFORMANCE CRITERIA	ELEMENTS	PROCESS ASSESSMENT	COMPETENCE ASSESSMENT		
			SERVICE ASSESSMENT	**UNDERPINNING KNOWLEDGE ASSESSMENT**	**TRAINING REQUIREMENTS**
				• The importance of creating charts **Circumstantial Knowledge:** **Detailed knowledge about:** • Confidentiality, trustworthy and security. • Safety precautions in handling computer system.	

LEVEL: II

MODULE: 1 MODULE TITLE: MAINTAINING NETWORK AND ITS COMPONENTS [DURATION: 141 HOURS]

UNIT NO. 1 UNIT TITLE: CABLING NETWORK DURATION: 90 HOURS

PERFORMANCE CRITERIA	ELEMENTS	PROCESS ASSESSMENT	COMPETENCE ASSESSMENT		
			SERVICE ASSESSMENT	**UNDERPINNING KNOWLEDGE ASSESSMENT**	**TRAINING REQUIREMENTS**
A person who has achieved this unit should be able to interpret and install passive network devices according to network layout map.	1. Install the cabling. 2. Updating of cabling. 3. Troubleshooting cabling	**The trainee should be able to:** • Drill • Install trunks • Install cable • Terminate cable • Install patch panel	The passive network installed and working as per networking standards.	**Knowledge evidence:** **Detailed Knowledge of:** **Method used:** The trainee should explain the different methods of installing network cables and patch panels. **Principles:** The trainee should explain the principals involved in installation of passive network. **Theories:** The trainee should explain: • The importance color codes. • Different network standards • The functions of different passive devices.	This unit can be achieved at work place or at training institution. The following tools and equipment should be available: • Clipping tool • Cable Tester • Set of screw drivers • Cable cutter • Drilling machine • Ball pen hammer • Dust Mask

PERFORMANCE CRITERIA	ELEMENTS	PROCESS ASSESSMENT	COMPETENCE ASSESSMENT		
			SERVICE ASSESSMENT	**UNDERPINNING KNOWLEDGE ASSESSMENT**	**TRAINING REQUIREMENTS**
				Circumstantial Knowledge: **Detailed knowledge about:** • Confidentiality, trustworthy and security. • Safety precautions in handling network system.	

LEVEL: II

MODULE: 1 MODULE TITLE: MAINTAINING NETWORK AND ITS COMPONENTS

UNIT NO. 2 UNIT TITLE: TRUNKING NETWORK DURATION: 51 HOURS

PERFORMANCE CRITERIA	ELEMENTS	PROCESS ASSESSMENT	COMPETENCE ASSESSMENT		
			SERVICE ASSESSMENT	**UNDERPINNING KNOWLEDGE ASSESSMENT**	**TRAINING REQUIREMENTS**
A person who has achieved this unit should be able to interpret and install passive network devices according to network layout map.	1. Install the Trunking. 2. Updating of Trunking. 3. Troubleshooting Trunking	**The trainee should be able to:** • Drill • Install trunks	The passive network installed and working as per networking standards.	**Knowledge evidence:** **Detailed Knowledge of:** **Method used:** The trainee should explain the different methods of installing network cables and patch panels. **Principles:** The trainee should explain the principals involved in installation of passive network. **Theories:** The trainee should explain: • The importance of Trunking. • The functions of different passive devices.	This unit can be achieved at work place or at training institution. The following tools and equipment should be available: • Clipping tool • Cable Tester • Set of screw drivers • Cable cutter • Drilling machine • Ball pen hammer • Dust Mask

PERFORMANCE CRITERIA	ELEMENTS	PROCESS ASSESSMENT	COMPETENCE ASSESSMENT		
			SERVICE ASSESSMENT	UNDERPINNING KNOWLEDGE ASSESSMENT	TRAINING REQUIREMENTS
				Circumstantial Knowledge: **Detailed knowledge about:** • Confidentiality, trustworthy and security. • Safety precautions in handling network system.	

LEVEL: II

MODULE: 2 **MODULE TITLE: MAINTAINING NETWORK SOFTWARE** **[DURATION: 249 HOURS]**

UNIT NO. 1 **UNIT TITLE: INSTALLATION OF NETWORKING SOFTWARE** **DURATION: 99 HOURS**

PERFORMANCE CRITERIA	ELEMENTS	PROCESS ASSESSMENT	COMPETENCE ASSESSMENT		
			SERVICE ASSESSMENT	UNDERPINNING KNOWLEDGE ASSESSMENT	TRAINING REQUIREMENTS
A person who has achieved this unit should be able to interpret and install active devices according to network layout map.	1. Install the network software. 2. Sharing files. 3. Updating of networking software. 4. Troubleshooting of networking software.	**The trainee should be able to:** • Install network interface. • Configure network interface. • Install network switch. • Install network router.	Active network devices installed and working as per networking standards.	**Knowledge evidence:** **Detailed Knowledge of:** **Method used:** The trainee should explain the different methods of installing active network devices. **Principles:** The trainee should explain the principals involved in installing active network devices. **Theories:** The trainee should explain: • Functions of different active network devices • Different network standards.	This unit can be achieved at work place or at training institution. The following tools and equipment should be available: • Computer set. • A Computer hardware toolkit. • Network interface. • Network switch. • Network router.

PERFORMANCE CRITERIA	ELEMENTS	PROCESS ASSESSMENT	COMPETENCE ASSESSMENT		
			SERVICE ASSESSMENT	UNDERPINNING KNOWLEDGE ASSESSMENT	TRAINING REQUIREMENTS
				Circumstantial Knowledge: **Detailed knowledge about:** • Confidentiality, trustworthy and security. • Safety precautions in handling network system.	

LEVEL: II

MODULE: 2 **MODULE TITLE: MAINTAINING NETWORK SOFTWARE**

UNIT NO. 2 **UNIT TITLE: PERFORMING PERIODIC BACKUP** **DURATION: 50 HOURS**

PERFORMANCE CRITERIA	ELEMENTS	PROCESS ASSESSMENT	COMPETENCE ASSESSMENT		
			SERVICE ASSESSMENT	UNDERPINNING KNOWLEDGE ASSESSMENT	TRAINING REQUIREMENTS
A person who has achieved this unit should be able to perform periodic backup according to backup procedures.	1. Backup media 2. Types of Back up 3. Backup the System 4. Backup Application and data File	**The trainee should be able to:** • Formulate periods for data backups. • Connect the backup drive. • Backup data. • Restore data • Disconnect the backup drive.	The data backed up as per requirements.	**Knowledge evidence:** **Detailed Knowledge of:** **Method used:** The trainee should explain the different methods of data backup. **Principles:** The trainee should explain the principals involved in data backup. **Theories:** The trainee should explain: • The importance of data backup. • Different types of backup software and drives.	This unit can be achieved at work place or at training institution. The following tools and equipment should be available: • Computer set • Backup software • Backup drive • External storage devices

PERFORMANCE CRITERIA	ELEMENTS	PROCESS ASSESSMENT	COMPETENCE ASSESSMENT		
			SERVICE ASSESSMENT	UNDERPINNING KNOWLEDGE ASSESSMENT	TRAINING REQUIREMENTS
				Circumstantial Knowledge: **Detailed knowledge about:** • Confidentiality, Trustworthy and security. • Ensuring safety precaution during data recovery.	

LEVEL: II

MODULE: 2 **MODULE TITLE: MAINTAINING NETWORK SOFTWARE**

UNIT NO. 3 **UNIT TITLE: PERFORMING DATA RECOVERY** **DURATION: 50 HOURS**

<table>
<tr><th rowspan="2">PERFORMANCE CRITERIA</th><th rowspan="2">ELEMENTS</th><th rowspan="2">PROCESS ASSESSMENT</th><th colspan="3">COMPETENCE ASSESSMENT</th></tr>
<tr><th>SERVICE ASSESSMENT</th><th>UNDERPINNING KNOWLEDGE ASSESSMENT</th><th>TRAINING REQUIREMENTS</th></tr>
<tr><td>A person who has achieved this unit should be able to recover data according to data recovery software manual.</td><td>1. Recovering of System,
2. Recovering of Applications
3. Recovering of data files.</td><td>The trainee should be able to:
• Select recovery software to use
• Recover data
• Save recovered data.</td><td>Data recovered as per requirements.</td><td>Knowledge evidence:
Detailed Knowledge of:
Method used:
The trainee should explain the different methods of recovering data.
Principles:
The trainee should explain the principals involved in recovering data.
Theories:
The trainee should explain:
• The importance of data recovery.
• Functions of different software for data recovery.
Circumstantial</td><td>This unit can be achieved at work place or at training institution.
The following tools and equipment should be available:
• Computer set
• Recovery software
• External storage devices
• Disk managers</td></tr>
</table>

PERFORMANCE CRITERIA	ELEMENTS	PROCESS ASSESSMENT	COMPETENCE ASSESSMENT		
			SERVICE ASSESSMENT	UNDERPINNING KNOWLEDGE ASSESSMENT	TRAINING REQUIREMENTS
				Knowledge: **Detailed knowledge about:** • Confidentiality, Trustworthy and security • Ensuring safety precaution during data recovery.	

LEVEL: II

MODULE: 2 **MODULE TITLE: MAINTAINING NETWORK SOFTWARE**

UNIT NO. 4 **UNIT TITLE: INSTALLATION OF NETWORKING DRIVERS** **DURATION: 50 HRS**

PERFORMANCE CRITERIA	**ELEMENTS**	**PROCESS ASSESSMENT**	**COMPETENCE ASSESSMENT**		
			SERVICE ASSESSMENT	**UNDERPINNING KNOWLEDGE ASSESSMENT**	**TRAINING REQUIREMENTS**
A person who has achieved this unit should be able to perform a thoroughly installation of the network drivers as per maintenance schedules.	1. Install the networking drivers. 2. Updating of networking drivers. 3. Troubleshooting networking drivers.	**The trainee should be able to:** • Select software diagnostic program. • Check the reachability • Test the cable • Fix the problem • Test the network	The diagnosed and repaired network works as per standards.	**Knowledge evidence:** **Detailed Knowledge of:** **Method used:** The trainee should explain the different methods of diagnosing and repairing networks. **Principles:** The trainee should explain the principles involved in diagnosing and repairing networks. **Theories:** The trainee should explain: • Importance of diagnosing and repairing networks. • Functions of diagnostic tools	This unit can be achieved at work place or at training institution. The following tools and equipment should be available: • Computer set • Network cable tester • Diagnostic software tools • Clipping tool • Set of screw drivers • Cable cutter • Drilling machine • Ball pen hammer • Dust Mask

PERFORMANCE CRITERIA	ELEMENTS	PROCESS ASSESSMENT	COMPETENCE ASSESSMENT		
			SERVICE ASSESSMENT	**UNDERPINNING KNOWLEDGE ASSESSMENT**	**TRAINING REQUIREMENTS**
				Circumstantial Knowledge: **Detailed knowledge about:** • Confidentiality, trustworthy and security. • Safety precautions in handling computer system.	

LEVEL: II

MODULE: 3 **MODULE TITLE: MANAGING DATABASE AND PRESENTATION APPLICATIONS**

UNIT NO. 1 **UNIT TITLE: DATABASE AND PRESENTATION APPLICATIONS.**

PERFORMANCE CRITERIA	ELEMENTS	PROCESS ASSESSMENT	COMPETENCE ASSESSMENT		
			SERVICE ASSESSMENT	UNDERPINNING KNOWLEDGE ASSESSMENT	TRAINING REQUIREMENTS
A person who has achieved this unit should be able to insert and retrieve data from the database application programs according to data validation rules of the program.	1. Selecting and creating database. 2. Creating Table. 3. Creating Query. 4. Creating Form. 5. Creating Reports.	**The trainee should be able to:** • Switch ON the computer • Open the database application program • Enter data • Query data. • Create Form • Print reports.	The data entered and retrieved as per requirements.	**Detailed Knowledge of:** **Method used:** The trainee should explain different methods of entering and retrieving data. **Principles:** The trainee should explain the principals involved in entering and retrieving data. **Theories:** The trainee should explain: • The function and importance of using database application. • Different data formats • Different database application programs	This unit can be achieved at work place or at training institution. The following tools and equipment should be available: • Computer set • Database application program • External storage devices • Printer

PERFORMANCE CRITERIA	ELEMENTS	PROCESS ASSESSMENT	COMPETENCE ASSESSMENT		
			SERVICE ASSESSMENT	**UNDERPINNING KNOWLEDGE ASSESSMENT**	**TRAINING REQUIREMENTS**
				Circumstantial Knowledge: **Detailed knowledge about:** • Confidentiality, trustworthy and security of data. • Safety precautions in handling computer system.	

LEVEL: II

MODULE: 3 MODULE TITLE: MANAGING DATABASE AND PRESENTATION APPLICATIONS

UNIT NO. 2 UNIT TITLE: PRESENTATION APPLICATIONS

<table>
<tr><th rowspan="2">PERFORMANCE CRITERIA</th><th rowspan="2">ELEMENTS</th><th rowspan="2">PROCESS ASSESSMENT</th><th colspan="3">COMPETENCE ASSESSMENT</th></tr>
<tr><th>SERVICE ASSESSMENT</th><th>UNDERPINNING KNOWLEDGE ASSESSMENT</th><th>TRAINING REQUIREMENTS</th></tr>
<tr>
<td>A person who has achieved this unit should be able to design and develop different presentation according to the PowerPoint program procedures.</td>
<td>1. Creating Presentation.
2. Edit Presentation.
3. Use Presentation tools.</td>
<td>The trainee should be able to:
• Switch ON the computer.
• Open PowerPoint program.
• Select slide layout type.
• Select the slide design.
• Edit slides according to the designed type.
• Insert objects.
• Format the presentation.
• Save the presentation.
• Print presentation.
• View show of the presentation</td>
<td>The presentation developed according to requirements and PowerPoint standards.</td>
<td>Knowledge evidence:
Detailed Knowledge of:
Method used:
The trainee should explain different methods of creating different presentations.
Principles:
The trainee should explain the principals involved in creating presentations.
Theories:
The trainee should explain:
• The function and importance of using PowerPoint program.
• Different types of presentations.</td>
<td>This unit can be achieved at work place or at training institution.
The following tools and equipment should be available:
• Computer set
• PowerPoint program
• Scanner
• Multimedia projector
• Multimedia projector screen
• External storage devices
• Black or Color Printer</td>
</tr>
</table>

<table>
<tr><th rowspan="2">PERFORMANCE CRITERIA</th><th rowspan="2">ELEMENTS</th><th rowspan="2">PROCESS ASSESSMENT</th><th colspan="3">COMPETENCE ASSESSMENT</th></tr>
<tr><th>SERVICE ASSESSMENT</th><th>UNDERPINNING KNOWLEDGE ASSESSMENT</th><th>TRAINING REQUIREMENTS</th></tr>
<tr><td></td><td></td><td></td><td></td><td>Circumstantial Knowledge:
Detailed knowledge about:
• Confidentiality and Trustworthy and Security.
• Safety precautions in handling computer system.</td><td></td></tr>
</table>

LEVEL: III

MODULE NUMBER: 1 **MODULE TITLE: MANAGING SAFE WORK ENVIRONMENT**

UNIT NO: 1 **UNIT TITLE: MANAGING HAZARDS** **DURATION: 43 hours.**

<table>
<tr><th rowspan="2">PERFORMANCE CRITERIA</th><th rowspan="2">ELEMENTS</th><th colspan="3">COMPETENCE ASSESSMENT</th><th rowspan="2">TRAINING REQUIREMENTS</th></tr>
<tr><th>PROCESS ASSESSMENT</th><th>SERVICE ASSESSMENT</th><th>UNDERPINNING KNOWLEDGE ASSESSMENT</th></tr>
<tr><td>A person who has achieved this unit should be able to manage work place hazards according to OSHA rules and regulations.</td><td>1. Control mechanical hazards.
2. Control chemical hazards.
3. Control physical hazards.</td><td>The trainee should be able to:
• Select tools and equipment
• Use OSHA rules and regulations
• Prepare workshop inspection report
• Identify any safety hazard materials
• Handle hazards material
• Identify and apply all emergency equipment and supplies
• Conduct safety awareness training</td><td>Hazards, risks, incident and accidents are managed according to OSHA's rules and regulations.</td><td>Knowledge evidence:
Detailed knowledge of:
Method used: The trainee should explain how to:
• Interpret OSHA rules and regulations
• Use safety gears
• Prepare preventive maintenance schedule and inspection report
• Prepare warning signs and safety instructions
• Conduct assessment
• Carry out accident investigation
• Monitor safe working environment</td><td>This unit can be achieved at a work place or in a training institution.
The following tools, equipment and safety gears should be available:-
• Gemological equipment
• Lapidary equipment
• First aid kit
• Fire extinguishers
• OSHA rules and regulations
• Helmet
• Hand loves
• Ear plug
• Dust mask</td></tr>
</table>

PERFORMANCE CRITERIA	ELEMENTS	COMPETENCE ASSESSMENT			TRAINING REQUIREMENTS
		PROCESS ASSESSMENT	**SERVICE ASSESSMENT**	**UNDERPINNING KNOWLEDGE ASSESSMENT**	
		to sub-ordinates • Monitor safety environment • Manage uses of safety gears • Cleaning tools and equipment • Storing tools and equipment		• Manage uses of safety gears **Principles:** The trainee should explain the principles of: • Preparing inspection check lists • Preparing warning signs and safety instructions • Identifying hazards materials • Preparing and conducting training • Handing hazard materials	

PERFORMANCE CRITERIA	ELEMENTS	COMPETENCE ASSESSMENT			TRAINING REQUIREMENTS
		PROCESS ASSESSMENT	**SERVICE ASSESSMENT**	**UNDERPINNING KNOWLEDGE ASSESSMENT**	
				Theories: The trainee should explain: • Function of inspection check list • Importance of posting warning sign and safety instructions • Advantages of risk assessment • Importance of carry out accident investigation • Importance of monitor safety at working place	

PERFORMANCE CRITERIA	ELEMENTS	COMPETENCE ASSESSMENT			TRAINING REQUIREMENTS
		PROCESS ASSESSMENT	SERVICE ASSESSMENT	UNDERPINNING KNOWLEDGE ASSESSMENT	
				Circumstantial knowledge: **Detailed knowledge about:** • Safety precautions while manage hazards • Safe handling of tools and equipment • Waste disposal	

LEVEL: III

MODULE NUMBER: 1 **MODULE TITLE: MANAGING SAFE WORK ENVIRONMENT [DURATION: 130 hours.]**

UNIT NO: 2 **UNIT TITLE: CARRYING OUT RISK ASSESSMENT** **DURATION: 43 hours.**

PERFORMANCE CRITERIA	ELEMENTS	COMPETENCE ASSESSMENT			TRAINING REQUIREMENTS
		PROCESS ASSESSMENT	SERVICE ASSESSMENT	UNDERPINNING KNOWLEDGE ASSESSMENT	
A person who has achieved this unit should be able to carry out risk assessment as per OSHA standards and regulations.	1. Control risk. 2. Manage safety gears. 3. Managing workshop safety rules.	**The trainee should be able to:** • Select tools and equipment • Supervise practice safe workshop practices to protect yourself, others and properties • React correctly and safely when faced with an emergency • Identify and apply correctly all emergency equipment and supplies	Risk assessment carried out as per OSHA standard and regulations.	**Knowledge evidence:** **Detailed knowledge of:** **Method used:** The trainee should explain how to: • Conduct safety training • Identify safety hazard material • Handle hazard material • Prepare inspection report **Principles:** The trainee should explain the principles of: • Reacting correctly and safely when faced with an emergency	This unit can be achieved at a work place or in a training institution. The following tools, equipment and safety gears should be available:- • OSHA regulations • Workshop rules • Risk assessment sheet • Dust mask • Ear plug • Hand gloves • Overall • Safety boots • Safety clear glasses

PERFORMANCE CRITERIA	ELEMENTS	COMPETENCE ASSESSMENT			TRAINING REQUIREMENTS
		PROCESS ASSESSMENT	**SERVICE ASSESSMENT**	**UNDERPINNING KNOWLEDGE ASSESSMENT**	
		• Make periodic inspections of workshop area and all equipment and prepare report • Conduct safety training • Identify any safety hazard material • Handle hazard material correctly • Make out and file safe report • Ensure availability of personal protective equipment • Monitor good environmental practices		• Identifying and applying correctly all emergency equipment and supplies • Conducting safety training • Identifying safely hazard materials • Handling hazard materials **Theories:** The trainee should explain: • Carrying out risk assessment • Conducting safety training • Inspecting workshop areas tools and equipment • Handling Hazard material correctly	

PERFORMANCE CRITERIA	ELEMENTS	COMPETENCE ASSESSMENT			TRAINING REQUIREMENTS
		PROCESS ASSESSMENT	SERVICE ASSESSMENT	UNDERPINNING KNOWLEDGE ASSESSMENT	
		• Clean tools and equipment • Store tools and equipment		**Circumstantial knowledge:** **Detailed knowledge about:** • Safety precautions while carrying out risk management • Safe handling of tools and equipment • Waste disposal	

LEVEL: III

MODULE NUMBER: 1 **MODULE TITLE: MANAGING SAFE WORK ENVIRONMENT**

UNIT NO: 3 **UNIT TITLE: MANAGING ENVIRONMENT** **DURATION: 43 hours.**

<table>
<tr><th rowspan="2">PERFORMANCE CRITERIA</th><th rowspan="2">ELEMENTS</th><th colspan="3">COMPETENCE ASSESSMENT</th><th rowspan="2">TRAINING REQUIREMENTS</th></tr>
<tr><th>PROCESS ASSESSMENT</th><th>PRODUCT ASSESSMENT</th><th>UNDERPINNING KNOWLEDGE ASSESSMENT</th></tr>
<tr><td>A person who has achieved this unit should be able to manage environment a per OSHA rules and regulations.</td><td>1. Managing air pollution.
2. Managing water pollution.
3. Managing land pollution.</td><td>The trainee should be able to:
• Select relevant safety gears
• Prepare preventive maintenance schedule
• Control environmental pollution
• Maintaining safety environment
• Managing safety personal environment
• Control tools, equipment and safety gears

• Control different</td><td>Workshop environment management as per rules and regulations.</td><td>Knowledge evidence:
Detailed knowledge of:
Method used: The trainee should explain how to:
• Interpret OSHA rules and regulations
• Prepare preventive maintenance schedule and inspection report
• Monitor safe working environment
• Control environment pollution
• Control different types of wastes
• Manage uses of safety gears

Principles: The trainee</td><td>This unit can be achieved at a work place or training institution.
The following tools, equipment and safety gears should be available:-
• Tool kit
• Sprit level
• Safety boots
• Hand gloves
• Overalls
• Hoe
• Broom
• Brush
• Dust covers
• Dust mask
• Dust bin
• Wheel barrow
• Slashes
• racks</td></tr>
</table>

PERFORMANCE CRITERIA	ELEMENTS	COMPETENCE ASSESSMENT			TRAINING REQUIREMENTS
		PROCESS ASSESSMENT	**PRODUCT ASSESSMENT**	**UNDERPINNING KNOWLEDGE ASSESSMENT**	
		types of wastes as per OSHA guideline • Conduct safety training awareness to subordinates • Clean tools and equipment • Store tools and equipment		should explain the principles of: • Managing environment pollution • Handling environmental safety work • Preparing and conducting safety training • Handling different types of wastes **Theories:** The trainee should explain: • Importance of safe work environment • Explain types of environmental pollution • Advantages of monitoring environmental pollution • Importance of preparing environmental schedule	

PERFORMANCE CRITERIA	ELEMENTS	COMPETENCE ASSESSMENT			TRAINING REQUIREMENTS
		PROCESS ASSESSMENT	PRODUCT ASSESSMENT	UNDERPINNING KNOWLEDGE ASSESSMENT	
				• Importance of control different types of wastes **Circumstantial knowledge:** **Detailed knowledge about:** • Safety knowledge while managing environmental pollution • Safe handling of tools and equipment • Waste disposal	

LEVEL: III

MODULE: 2 **MODULE TITLE: MAINTAINING WEBSITE**

UNIT NO. 1 **UNIT TITLE: WEBSITE DESIGN LAYOUTS** **DURATION: 20 hours.**

PERFORMANCE CRITERIA	ELEMENTS	PROCESS ASSESSMENT	COMPETENCE ASSESSMENT		
			SERVICE ASSESSMENT	**UNDERPINNING KNOWLEDGE ASSESSMENT**	**TRAINING REQUIREMENTS**
A person who has achieved this unit should be able to design a simple functional website per principles of website designing.	1. Creating Page Layouts. 2. Using Website Templates.	**The trainee should be able to:** • Switch ON the computer • Open the Website designing program • Design website layout • Design the website map. • Choose colour and fonts	The website designed as per requirement of specific target user group	**Knowledge evidence:** **Detailed knowledge of:** **Method used:** The trainee should explain different methods of designing effective website. **Principles:** The trainee should be able to explain the principals involved in designing effective website. **Theories:** The trainee should explain: • The strengths and weaknesses of major websites. • The reasons to the choice of fonts and colors that suit specific website **Circumstantial Knowledge:** **Detailed knowledge**	This unit can be achieved at work place or at training institution. The following tools and equipment should be available: • Computer set • Website designing tools

PERFORMANCE CRITERIA	ELEMENTS	PROCESS ASSESSMENT	COMPETENCE ASSESSMENT		
			SERVICE ASSESSMENT	UNDERPINNING KNOWLEDGE ASSESSMENT	TRAINING REQUIREMENTS
				about: • Confidentiality, Trustworthy and security • Plagiarism and website piracy. • Safety precautions in handling computer system.	

LEVEL: III

MODULE:2 **MODULE TITLE: MAINTAINING WEBSITE**

UNIT NO.2 **UNIT TITLE: CREATING WEB PAGES** **DURATION: 70 hours.**

PERFORMANCE	ELEMENTS	PROCESS ASSESSMENT	COMPETENCE ASSESSMENT

CRITERIA			SERVICE ASSESSMENT	UNDERPINNING KNOWLEDGE ASSESSMENT	TRAINING REQUIREMENTS
A person who has achieved this unit should be able to create a simple pages and appropriate page elements according to the procedures of website editing programs.	1. Using Website Designing Languages.	**The trainee should be able to:** • Switch ON the computer • Open the Website editing program • Create a new page • Format the page according to the designed layout • Insert the contents • Scan the images • Import or insert objects/images • Create links • Save the page	The web page created and formatted as per requirements.	**Knowledge evidence:** **Detailed knowledge of:** **Method used:** The trainee should explain different methods of creating web pages. **Principles:** The trainee should explain the principles involved in creating web pages. **Theories:** The trainee should explain: • The strengths and weaknesses of major tools used in creating web pages. • The functions and importance of web pages. • Different types of web pages. **Circumstantial Knowledge:**	This unit can be achieved at work place or at training institution. The following tools and equipment should be available: • Computer set • Scanner • Website editing tools

PERFORMANCE CRITERIA	ELEMENTS	PROCESS ASSESSMENT	COMPETENCE ASSESSMENT		
			SERVICE ASSESSMENT	**UNDERPINNING KNOWLEDGE ASSESSMENT**	**TRAINING REQUIREMENTS**
				Detailed knowledge about: • Confidentiality, Trustworthy and security • Plagiarism and website piracy. • Safety precautions in handling computer system.	

LEVEL: III

MODULE: 2 **MODULE TITLE: MAINTAINING WEBSITES**

UNIT NO. 3 **UNIT TITLE: UPLOADING CONTENTS** **DURATION: 20 hours.**

PERFORMANCE CRITERIA	ELEMENTS	PROCESS ASSESSMENT	COMPETENCE ASSESSMENT		
			SERVICE ASSESSMENT	UNDERPINNING KNOWLEDGE ASSESSMENT	TRAINING REQUIREMENTS
A person who has achieved this unit should be able to upload the website to the web server according to website map.	1. Using Uploading Software.	**The trainee should be able to**: • Switch ON the computer • Open the Website uploading program • Locate the website folder • Connect to the web-hosting server • Transfer files to web-hosting server • Open the web browser • Test the uploaded website	The uploaded website functions as per website map.	**Knowledge evidence:** **Detailed knowledge of:** **Method used:** The trainee should explain different methods of transferring files to the web-hosting server. **Principles:** The trainee should explain the principles involved in uploading the website contents. **Theories:** The trainee should explain: • The strengths and weaknesses of major uploading tools. • Different types of uploading tools.	This unit can be achieved at work place or at training institution. The following tools and equipment should be available: • Computer set • Website uploading tools • Internet connection • Web hosting server

PERFORMANCE CRITERIA	ELEMENTS	PROCESS ASSESSMENT	COMPETENCE ASSESSMENT		
			SERVICE ASSESSMENT	UNDERPINNING KNOWLEDGE ASSESSMENT	TRAINING REQUIREMENTS
				Circumstantial Knowledge: **Detailed knowledge about:** • Confidentiality, Trustworthy and security • Security of the contents • Plagiarism and website piracy. • Safety precautions in handling computer system.	

LEVEL: III

MODULE: 2 **MODULE TITLE: MAINTAINING WEBSITES**

UNIT NO. 4 **UNIT TITLE: MAINTAINING WEBSITE AND WEB HOSTING SERVER DURATION: 20 hours.**

PERFORMANCE CRITERIA	ELEMENTS	PROCESS ASSESSMENT	COMPETENCE ASSESSMENT		
			SERVICE ASSESSMENT	UNDERPINNING KNOWLEDGE ASSESSMENT	TRAINING REQUIREMENTS
A person who has achieved this unit should be able to maintain the website and web hosting server according to web maintenance application interface.	Using Website Maintenance Software.	**The trainee should be able to:** • Switch ON the computer • Open web browser • Connect to the server • Open web maintenance program • Update the content and services of the web hosting server • Save the changes • Disconnect from the server	The updated changes of the website and web hosting server functions as per requirements.	**Knowledge evidence:** **Detailed knowledge of:** **Method used:** The trainee should explain different methods of updating web contents and services to the web-hosting server. **Principles:** The trainee should explain the principles involved in maintaining website and web-hosting server. **Theories:** The trainee should explain: • Different tools used to maintain websites and web hosting server • The strengths and	This unit can be achieved at work place or at training institution. The following tools and equipment should be available: • Computer set • Internet connection • Web maintenance tools • Web hosting server

PERFORMANCE CRITERIA	ELEMENTS	PROCESS ASSESSMENT	COMPETENCE ASSESSMENT		
			SERVICE ASSESSMENT	UNDERPINNING KNOWLEDGE ASSESSMENT	TRAINING REQUIREMENTS
				weaknesses of major web-hosting servers. **Circumstantial Knowledge:** **Detailed knowledge about:** • Confidentiality and trustworthy • Security of the website and the web-hosting server. • Safety precautions in handling computer system.	

LEVEL: III

MODULE: 3 **MODULE TITLE: DESIGNING GRAPHICS** **[DURATION: 150 hours.]**

UNIT NO. 1 **UNIT TITLE: SKETCHING GRAPHICS** **DURATION: 55 hours.**

PERFORMANCE CRITERIA	ELEMENTS	PROCESS ASSESSMENT	COMPETENCE ASSESSMENT		
			SERVICE ASSESSMENT	**UNDERPINNING KNOWLEDGE ASSESSMENT**	**TRAINING REQUIREMENTS**
A person who has achieved this unit should be able to design and create graphics sketches according to fine art procedures.	1. Identify tools for Sketching 2. Draw Shapes. 3. Grouping. 4. Editing Shapes. 5. Colouring	**The trainee should be able to:** • Perform measurements • Draw different shapes • Combine the shapes • Edit the combined shapes to form a sketch • Paint the sketch	The created sketch drawn and painted as per standards of fine art.	**Knowledge evidence:** **Detailed knowledge of:** **Method used:** The trainee should describe different methods of creating sketches. **Principles:** The trainee should explain principles involved in creating sketches. **Theories:** The trainee should explain: • Why is it necessary to create sketches before creating graphics? • Types of material used in creating sketches.	This unit can be achieved at work place or at training institution. The following tools and equipment should be available: • Drawing pencils • Rubber • Plain papers • Set of rulers • Compass • Color set

PERFORMANCE CRITERIA	ELEMENTS	PROCESS ASSESSMENT	COMPETENCE ASSESSMENT		
			SERVICE ASSESSMENT	**UNDERPINNING KNOWLEDGE ASSESSMENT**	**TRAINING REQUIREMENTS**
				Circumstantial Knowledge: **Detailed knowledge about:** • Confidentiality, Trustworthy and Security • Plagiarism and copyright.	

LEVEL: III

MODULE: 3 **MODULE TITLE: DESIGNING GRAPHICS**

UNIT NO. 2 **UNIT TITLE: WORK WITH GRAPHICS** **DURATION: 65 hours.**

PERFORMANCE CRITERIA	ELEMENTS	PROCESS ASSESSMENT	COMPETENCE ASSESSMENT		
			SERVICE ASSESSMENT	UNDERPINNING KNOWLEDGE ASSESSMENT	TRAINING REQUIREMENTS
A person who has achieved this unit should be able to create graphics according to graphics editing program procedures.	1. Creating Graphics 2. Editing graphics.	**The trainee should be able to:** • Choose graphic editing program • Scan sketch • Import the scanned sketch into graphics editing tool • Edit the sketch • Create graphic • Save the graphics	The graphic created as per graphics design standards.	**Knowledge evidence:** **Detailed knowledge of:** **Method used:** The trainee should describe different methods of creating graphics. **Principles:** The trainee should explain the principles involved in creating graphics. **Theories:** The trainee should explain: • Functions and importance of graphics. • Identify the strengths and weaknesses of major graphic editing tools. • Different graphics editing tools. **Circumstantial**	This unit can be achieved at work place or at training institution. The following tools and equipment should be available: • Computer set • Graphics editing program • Printer • External storage devices • Scanner

PERFORMANCE CRITERIA	ELEMENTS	PROCESS ASSESSMENT	COMPETENCE ASSESSMENT		
			SERVICE ASSESSMENT	**UNDERPINNING KNOWLEDGE ASSESSMENT**	**TRAINING REQUIREMENTS**
				Knowledge: **Detailed knowledge about:** • Confidentiality, Trustworthy and Security • Plagiarism and copyright.	

LEVEL: III

MODULE: 3 **MODULE TITLE: DESIGNING GRAPHICS**

UNIT NO.3 **UNIT TITLE: COLOUR MIXING** **DURATION: 30 hours.**

PERFORMANCE CRITERIA	ELEMENTS	PROCESS ASSESSMENT	COMPETENCE ASSESSMENT		
			SERVICE ASSESSMENT	UNDERPINNING KNOWLEDGE ASSESSMENT	TRAINING REQUIREMENTS
A person who has achieved this unit should be able to mix color according to color mixing rule.	1. Applying Colours 2. Edit color	**The trainee should be able to:** • Choose graphic editing program • Scan sketch • Import the scanned sketch into graphics editing tool • Edit the sketch • Create graphic • Save the graphics	The color combined according to the color combination principles.	**Knowledge evidence:** **Detailed knowledge of:** **Method used:** The trainee should explain the process of color mixing **Principles:** The trainee should explain the principals involved in color mixing. **Theories:** The trainee should explain: • The importance of differentiating color. • The importance of primary and secondary color.	This unit can be achieved at work place or at training institution. The following tools and equipment should be available: • Computer set • Graphics editing tool

PERFORMANCE CRITERIA	ELEMENTS	PROCESS ASSESSMENT	COMPETENCE ASSESSMENT		
			SERVICE ASSESSMENT	UNDERPINNING KNOWLEDGE ASSESSMENT	TRAINING REQUIREMENTS
				Circumstantial Knowledge: **Detailed knowledge about:** • Confidentiality, Trustworthy and Security. • Safety precautions in handling computer system.	

LEVEL: III

MODULE: 4 **MODULE TITLE: MANAGING INTERNET SERVICES** **[DURATION: 80 hours.]**
UNIT NO. 1 **UNIT TITLE: SEARCHING INFORMATION** **DURATION: 25 hours.**

PERFORMANCE CRITERIA	ELEMENTS	PROCESS ASSESSMENT	COMPETENCE ASSESSMENT		
			SERVICE ASSESSMENT	UNDERPINNING KNOWLEDGE ASSESSMENT	TRAINING REQUIREMENTS
A person who has achieved this unit should be able to search information from the Internet according to the procedures of searching engine.	1. Search world wide web (www) 2. Search Local Files	**The trainee should be able to:** • Switch ON the computer • Establish Internet connection • Launch browser • Navigate to search engine • Select Indexed categories • Create search key • Submit search key to search engine • Sort the search results • Save or print the results	The information obtained as per requirements.	**Knowledge evidence:** **Detailed knowledge of:** **Method used:** The trainee should explain the different ways of searching information. **Principles:** The trainee should explain the principles involved in searching information. **Theories:** The trainee should explain: • The importance of using search engines. • Different search engines. • Why search key is important	This unit can be achieved at work place or at training institution. The following tools and equipment should be available: • Computer set • Internet Connection • Search engines • External storage devices • Printer • Web browser application

PERFORMANCE CRITERIA	ELEMENTS	PROCESS ASSESSMENT	COMPETENCE ASSESSMENT		
			SERVICE ASSESSMENT	UNDERPINNING KNOWLEDGE ASSESSMENT	TRAINING REQUIREMENTS
				Circumstantial Knowledge: **Detailed knowledge about:** • Confidentiality, Trustworthy and security • Safety precautions in handling computer system.	

LEVEL: III

MODULE: 4 MODULE TITLE: MANAGING INTERNET SERVICES

UNIT NO. 2 UNIT TITLE: TRANSFERRING FILES DURATION: 25 hours.

PERFORMANCE CRITERIA	ELEMENTS	PROCESS ASSESSMENT	COMPETENCE ASSESSMENT		
			SERVICE ASSESSMENT	UNDERPINNING KNOWLEDGE ASSESSMENT	TRAINING REQUIREMENTS
A person who has achieved this unit should be able to download and upload files in the Internet according to the procedures of file Transferring application program.	1. Upload Files 2. Download files	**The trainee should be able to:** • Switch ON the computer • Establish Internet connection • Open file transfer application • Enter source or destination address • Select the upload or download files • Transfer files	Files uploaded/downl oaded as per requirements	**Knowledge evidence:** **Detailed knowledge of:** **Method used:** T The trainee should explain different methods of uploading and downloading files. **Principles:** The trainee should explain the principles involved in uploading and downloading files. **Theories:** The trainee should explain: • Different tools for downloading and uploading files. • The importance of file	This unit can be achieved at work place or at training institution. The following tools and equipment should be available: • Computer set • Internet Connection • File Transfer Application • External storage devices

PERFORMANCE CRITERIA	ELEMENTS	PROCESS ASSESSMENT	COMPETENCE ASSESSMENT		
			SERVICE ASSESSMENT	UNDERPINNING KNOWLEDGE ASSESSMENT	TRAINING REQUIREMENTS
				transfer. **Circumstantial Knowledge:** **Detailed knowledge about:** • Confidentiality, trustworthy and security. • Safety precautions in handling computer system.	

LEVEL: III

MODULE: 4 **MODULE TITLE: MANAGING INTERNET SERVICES**

UNIT NO. 3 **UNIT TITLE: USING E – MAIL** **DURATION: 30 hours.**

PERFORMANCE CRITERIA	ELEMENTS	PROCESS ASSESSMENT	COMPETENCE ASSESSMENT		
			SERVICE ASSESSMENT	UNDERPINNING KNOWLEDGE ASSESSMENT	TRAINING REQUIREMENTS
A person who has achieved this unit should be able to create and operate email account according to email application program procedures.	1. Creating E-Mail Account 2. Receive E-Mails. 3. Sending E-Mails.	**The trainee should be able to:** • Switch ON the computer • Establish internet connection • Open email application program • Create email account • Login into email account • Create email message • Attach files • Send email • Receive email • Print email message • Manage email	The E-mail Accounts created and managed as per requirements	**Knowledge evidence:** **Detailed knowledge of:** **Method used:** The trainee should explain the different methods of creating and operating email accounts. **Principles:** The trainee should explain the principles involved in managing email accounts. **Theories:** The trainee should explain: • The importance and functions of email application program. • Different type of email application program • Different types of email accounts.	This unit can be achieved at work place or at training institution. The following tools and equipment should be available: • Computer set • Internet connection • Email application program • External storage devices • Printer

PERFORMANCE CRITERIA	ELEMENTS	PROCESS ASSESSMENT	COMPETENCE ASSESSMENT		
			SERVICE ASSESSMENT	UNDERPINNING KNOWLEDGE ASSESSMENT	TRAINING REQUIREMENTS
		account		**Circumstantial Knowledge:** **Detailed knowledge about:** • Confidentiality, trustworthy and security. • Safety precautions in handling computer system. • Internet popup messages	

LEVEL: III

MODULE: 5 **MODULE TITLE: BASIC PROGRAMMING** **[DURATION: 120 hours.]**

UNIT NO. 1 **UNIT TITLE: PERFORM APPLICATION PROGRAMS** **DURATION: 40 hours.**

PERFORMANCE CRITERIA	ELEMENTS	PROCESS ASSESSMENT	COMPETENCE ASSESSMENT		
			SERVICE ASSESSMENT	UNDERPINNING KNOWLEDGE ASSESSMENT	TRAINING REQUIREMENTS
A person who has achieved this unit should be able to create user-friendly database application programs according to the principles of programming.	1. Identify Environments Tools 2. Applying Structure& Flow 3. Creating Interface	**The trainee should be able to:** • Create form interface • Connect interface with the database • Code the forms interface • Create reports • Create queries use SQL • Link reports with the forms • Print the report • Create the package	The application program created with minimum errors and produces reports as per standard.	**Knowledge evidence:** **Detailed knowledge of:** **Method used:** TThe trainee should explain different methods of creating database application programs. **Principles:** The trainee should explain the principles involved in database application programs. **Theories:** The trainee should explain: The logical concepts required in database programming.	This unit can be achieved at work place or at training institution. The following tools and equipment should be available: • Computer set • Programming environment tools • Printer

PERFORMANCE CRITERIA	ELEMENTS	PROCESS ASSESSMENT	COMPETENCE ASSESSMENT		
			SERVICE ASSESSMENT	UNDERPINNING KNOWLEDGE ASSESSMENT	TRAINING REQUIREMENTS
				Circumstantial Knowledge: **Detailed knowledge about:** • Confidentiality, Trustworthy and Security • Software piracy and security measures. • Safety precautions in handling computer system.	

LEVEL: III

MODULE: 5 **MODULE TITLE: BASIC PROGRAMMING**

UNIT NO. 2 **UNIT TITLE: CREATE SIMPLE PROGRAMS** **DURATION: 80 HRS**

PERFORMANCE CRITERIA	ELEMENTS	PROCESS ASSESSMENT	COMPETENCE ASSESSMENT		
			SERVICE ASSESSMENT	UNDERPINNING KNOWLEDGE ASSESSMENT	TRAINING REQUIREMENTS
A person who has achieved this unit should be able to create a well-normalized database as per relational database management systems procedures.	1. Writing Code. *2.* Data Access	**The trainee should be able to:** • Create table • Create Multiple document • Build table relationships • Write procedures • Write functions • Call public procedure/function • Create macros • Create reports	The database created with minimum data redundant and produces the report as per requirements.	**Knowledge evidence:** **Detailed knowledge of:** **Method used:** The trainee should explain the process of creating database. **Principles:** The trainee should explain the principals involved in • Creating database. • Write Function and Procedure • Program structure	This unit can be achieved at work place or at training institution. The following tools and equipment should be available: • Computer set • Database application programs

PERFORMANCE CRITERIA	ELEMENTS	PROCESS ASSESSMENT	COMPETENCE ASSESSMENT		
			SERVICE ASSESSMENT	UNDERPINNING KNOWLEDGE ASSESSMENT	TRAINING REQUIREMENTS
				Theories: The trainee should explain: • Why is the database so important? • Why normalization should be performed prior to building table relationships. • Why there exist different database application environments.	

PERFORMANCE CRITERIA	ELEMENTS	PROCESS ASSESSMENT	COMPETENCE ASSESSMENT		
			SERVICE ASSESSMENT	UNDERPINNING KNOWLEDGE ASSESSMENT	TRAINING REQUIREMENTS
				Circumstantial Knowledge: **Detailed knowledge about:** • Types of Error • Object Properties • Variable and Data Type • Safety precautions in handling computer system.	

LEVEL: III

MODULE NUMBER: 6 **MODULE TITLE: MANAGING PREVENTIVE MAINTENANCE** **DURATION: 40 hours.**

UNIT NO: 1 **UNIT TITLE: PLANNING PREVENTIVE MAINTENANCE** **DURATION: 20 hours.**

PERFORMANCE CRITERIA	**ELEMENTS**	**COMPETENCE ASSESSMENT**			**TRAINING REQUIREMENTS**
		PROCESS ASSESSMENT	**PRODUCT ASSESSMENT**	**UNDERPINNING KNOWLEDGE ASSESSMENT**	
A person who has achieved this unit should be able to plan preventive maintenance as per workshop standards.	1. Prepare schedules of preventive maintenance of tools, machines and equipment. 2. Prepare inspection check list of tools, equipment and machines.	**The trainee should be able to:** • Interpret service manuals • Read and apply workshop rules and regulations • Select tools and equipment • Make periodic inspection of workshop area and all equipment. • Prepare workshop inspection report of tools and equipment	Preventive maintenance is planned as per workshop standards.	**Knowledge evidence:** **Detailed knowledge of:** **Method used:** The trainee should explain how to: • Prepare workshop inspection report • Prepare workshop safety sign • Plan and prepare workshop inventory • Plan and prepare preventive maintenance training	This unit can be achieved at work place or in a training institution. The following tools, equipment and safety gears should be available:- • Tool kit • Service manuals • Workshop tool and equipment • Workshop rules and regulations • Hand gloves • Overall • Safety boots • Safety clear glasses • Helmet • Nose mask • Ear plug

PERFORMANCE CRITERIA	ELEMENTS	COMPETENCE ASSESSMENT			TRAINING REQUIREMENTS
		PROCESS ASSESSMENT	**PRODUCT ASSESSMENT**	**UNDERPINNING KNOWLEDGE ASSESSMENT**	
		• Prepare preventive maintenance programs • Prepare workshop preventive maintenance schedule • Plan and Prepare workshop inventory • Clean tools and equipment • Store tools and equipment		**Principles:** The trainee should explain the principles of: • Preparing preventive maintenance schedule • Plan and prepare workshop inventory **Theories:** The trainee should explain: • Importance of interpreting service manuals • Importance of preparing workshop inspection and maintenance schedule reports • Importance of preparing maintenance training programs • Importance of Cleaning and storing tools and equipment	

PERFORMANCE CRITERIA	ELEMENTS	COMPETENCE ASSESSMENT			TRAINING REQUIREMENTS
		PROCESS ASSESSMENT	PRODUCT ASSESSMENT	UNDERPINNING KNOWLEDGE ASSESSMENT	
				Circumstantial knowledge: **Detailed knowledge about:** • Safety precautions while planning preventive maintenance • Safe handling of tools and equipment • Waste disposal	

LEVEL: III

MODULE NUMBER: 6 **MODULE TITLE: MANAGING PREVENTIVE MAINTENANCE**

UNIT NO: 2 **UNIT TITLE: SUPERVISING PREVENTIVE MAINTENANCE** **DURATION: 20 hours.**

PERFORMANCE CRITERIA	ELEMENTS	COMPETENCE ASSESSMENT			TRAINING REQUIREMENTS
		PROCESS ASSESSMENT	PRODUCT ASSESSMENT	UNDERPINNING KNOWLEDGE ASSESSMENT	
A person who has achieved this unit should be able to plan preventive maintenance as per workshop standards.	1. Perform preventive maintenance of tools, equipment and machines. 2. Perform preventive maintenance of working environment.	**The trainee should be able to:** • Interpret service manuals • Read and apply rules and regulations • Prepare and apply workshop inspection report • Prepare and apply workshop preventive maintenance schedule • Plan and conduct preventive maintenance	Preventive maintenance of tools, equipment and machines are performed as per workshop standards.	**Knowledge evidence:** **Detailed knowledge of:** **Method used:** The trainee should explain how to: • Prepare and apply workshop preventive schedule • Plan and conduct preventive maintenance training • Correct hand tools and equipment safety • Follow good environmental practices	This unit can be achieved at work place or in a training institution. The following tools, equipment and safety gears should be available:- • Tool kit • Service manuals • Workshop tool and equipment • Workshop rules and regulations • Hand gloves • Overall • Safety boots • Safety clear glasses • Helmet • Nose mask

PERFORMANCE CRITERIA	ELEMENTS	COMPETENCE ASSESSMENT			TRAINING REQUIREMENTS
		PROCESS ASSESSMENT	PRODUCT ASSESSMENT	UNDERPINNING KNOWLEDGE ASSESSMENT	
		training • Practice correct hand tools and equipment safety • Monitor good and environmental practices • Clean tools and equipment • Store tools and equipment		**Principles:** The trainee should explain the principles of: • Preparing and applying preventive maintenance schedule • Plan and conduct preventive maintenance training **Theories:** The trainee should explain: • Importance of preparing and applying preventive maintenance schedule • Importance of Planning and conducting preventive maintenance training • Importance of follow good environmental practices	• Ear plug

PERFORMANCE CRITERIA	ELEMENTS	COMPETENCE ASSESSMENT			TRAINING REQUIREMENTS
		PROCESS ASSESSMENT	PRODUCT ASSESSMENT	UNDERPINNING KNOWLEDGE ASSESSMENT	
				Circumstantial knowledge: **Detailed knowledge about:** • Safety precautions while planning preventive maintenance • Safe handling of tools and equipment • Waste disposal	

LEVEL: III

MODULE NUMBER: 7 **MODULE TITLE: MANAGING COMPUTER ACTIVITIES** **DURATION: 80 hours.**

UNIT NO: 1 **UNIT TITLE: DESIGNING WORKSHOP LAYOUT** **DURATION: 10 HRS**

PERFORMANCE CRITERIA	ELEMENTS	COMPETENCE ASSESSMENT			TRAINING REQUIREMENTS
		PROCESS ASSESSMENT	PRODUCT ASSESSMENT	UNDERPINNING KNOWLEDGE ASSESSMENT	
A person who has achieved this unit should be able to design lab/workshop layout as per ministry of labor rules and regulations.	1. Designing preparation room 2. Designing layout for equipment	**The trainee should be able to:** • Select tools and equipment • Plan computer lab/workshop layout • Locate different workshop sections • Locate isles • Locate the installation of different machines • Identify places for safety gears equipment • Identify convenient place for stores • Identify convenient place to assemble in	Designed laboratory layout conforms to environmental and ministry of labour rules and regulations.	**Knowledge evidence:** **Detailed knowledge of:** **Method used:** The trainee should explain how to arrange different laboratory sections. **Principles:** The trainee should explain the principles of: • Laying out laboratory • Equipment installation in a laboratory	This unit can be achieved at a work place or training institution. The following equipment, tools and safety gears and should be available: • Organization structures • Different Laboratory layouts • Overhead projector • Computer. • Flip charts • Chalk board • Different management text books • Handouts • Stationery • Drawing instruments

PERFORMANCE CRITERIA	ELEMENTS	COMPETENCE ASSESSMENT			TRAINING REQUIREMENTS
		PROCESS ASSESSMENT	PRODUCT ASSESSMENT	UNDERPINNING KNOWLEDGE ASSESSMENT	
		case of emergency. • Mark emergency exit • Locate information resource centre • Locate laundry and latrines • Design security system of tools and equipment • Design safety system to workers • Identify marks and postures • Place sign mark and postures • Label safety precautions for laboratory materials and goods		**Theories:** The trainee should explain: • Steps to design laboratory layout • Components applied in laboratory safety and security systems **Circumstantial knowledge:** **Detailed knowledge about:** • Safety handling of working tools and equipment. • Waste disposal.	• Landline telephone • Balances • Fume cupboards • Oven • Water bath • Water distilling machines • Water de-ionizing machines • Fire extinguishing equipment • Cup boards • Drawers • Work benches • Shelves • Workshop coat • Rubber gloves • Safety boots • Masks • Safety clear goggles • Helmets

LEVEL III

MODULE NUMBER: 6 **MODULE TITLE MANAGING COMPUTER ACTIVITIES**

UNIT NO: 2 **UNIT TITLE: CONTROLLING TOOLS AND EQUIPMENT** **DURATION: 10 HRS**

PERFORMANCE CRITERIA	**ELEMENTS**	**COMPETENCE ASSESSMENT**			**TRAINING REQUIREMENTS**
		PROCESS ASSESSMENT	**SERVICE ASSESSMENT**	**UNDERPINNING KNOWLEDGE ASSESSMENT**	
A person who has achieved this unit should be able to control tools and equipment in laboratory as per regulations and PPRA guidelines.	1. Maintaining tools control system. 2. Taking inventory of tools and equipment	**The trainee should be able to:** • Select tools and equipment • Design tools storage system • Keep record of tools and equipment in laboratory • Record tools and equipment issued daily from stores • Record tools and equipment received daily from user • Record damaged tools and equipment • Record lost equipment and tools	Tools and equipment controlled as per Public Procurement Regulatory Act (PPRA) guidelines.	**Knowledge evidence:** **Detailed knowledge of:** **Method used:** The trainee should explain how to: • Maintain tools ledgers • Maintain Tools inventory records **Principles:** The trainee should explain the principles of: • Controlling tools and equipment in the laboratory • Record keeping in the laboratory	This unit can be achieved at a work place or training institution. The following equipment, tools and safety gears should be available: • Skills logbook • Tools and equipment catalogue • Stationeries • Scientific calculator • Staple machine • Binding machine • Tools list • Walldrobe • Bench with tool crip • Toolboxes • Tools issue voucher • Tools ledger

PERFORMANCE CRITERIA	ELEMENTS	COMPETENCE ASSESSMENT			TRAINING REQUIREMENTS
		PROCESS ASSESSMENT	SERVICE ASSESSMENT	UNDERPINNING KNOWLEDGE ASSESSMENT	
		• Discard damaged tools and equipment • Order new tools and equipment		**Theories:** The trainee should explain: • Properties of tools and equipment • Effects of weather on different tools • Necessary security on stores/workshops • Public Procurement Regulatory Act guidelines **Circumstantial knowledge:** **Detailed knowledge about:** • Safety precautions while controlling tools and equipment • Safe handling of tools and equipment • Waste disposal	• Equipment ledger • Tools inventory list • Files • Overcoat • Safety boots

LEVEL III

MODULE NUMBER: 6 **MODULE TITLE: MANAGING COMPUTER ACTIVITIES**

UNIT NO: 3 **UNIT TITLE: ESTIMATING MATERIAL AND LABOUR COST** **DURATION: 10 HRS**

<table>
<tr><th rowspan="2">PERFORMANCE CRITERIA</th><th rowspan="2">ELEMENTS</th><th colspan="3">COMPETENCE ASSESSMENT</th><th rowspan="2">TRAINING REQUIREMENTS</th></tr>
<tr><th>PROCESS ASSESSMENT</th><th>SERVICE ASSESSMENT</th><th>UNDERPINNING KNOWLEDGE ASSESSMENT</th></tr>
<tr><td>A person who has achieved this unit should be able to perform job costing as per job performed.</td><td>1. Maintaining records of Laboratory materials
2. Maintaining man hours /day of laboratory staff
3. Performing job cost calculations</td><td>The trainee should be able to:
• Select tools and equipment
• Read inspection report
• Prepare costs
• Prepare material cost estimates
• Prepare labour costs
• Prepare overhead costs
• Prepare material request
• Prepare quotations and distribute into various shops</td><td>Estimation of materials and labour costs performed as per job requirements.</td><td>Knowledge evidence:
Detailed knowledge of:
Method used: The trainee should explain how to calculate the costs of materials and labour.

Principles: The trainee should explain the principles of:
• Determining man hour rate to make job costing estimates.
• Calculating cost of materials and labour.</td><td>This unit can be achieved at a work place or training institution.
The following tools, equipment and safety gears should be available:
• List of spares parts
• Materials list
• Local purchasing order (LPO)
• Calculator/Computer
• Stationeries
• Binding machine
• Material requisition form or (Material requisition voucher (MRV)
• Job card
• Price list</td></tr>
</table>

PERFORMANCE CRITERIA	ELEMENTS	COMPETENCE ASSESSMENT			TRAINING REQUIREMENTS
		PROCESS ASSESSMENT	**SERVICE ASSESSMENT**	**UNDERPINNING KNOWLEDGE ASSESSMENT**	
		• Obtain proforma invoice from different shops • Prepare job costing including other overheads		**Theories:** The trainee should explain: • Importance of estimating materials and labour cost • Importance of using genuine materials • Use of parts catalogue **Circumstantial knowledge:** **Detailed knowledge about** safe handling of materials and documents	• Good received note (GRN) • Gloves • Overcoat • Safety boot • Mask

LEVEL III

MODULE NUMBER: 7 **MODULE TITLE: MANAGING COMPUTER ACTIVITIES**

UNIT NO: 4 **UNIT TITLE: TRAINING SUB-ORDINATES** **DURATION: 15 HRS**

<table>
<tr><th rowspan="2">PERFORMANCE CRITERIA</th><th rowspan="2">ELEMENTS</th><th colspan="3">COMPETENCE ASSESSMENT</th><th rowspan="2">TRAINING REQUIREMENTS</th></tr>
<tr><th>PROCESS ASSESSMENT</th><th>SERVICE ASSESSMENT</th><th>UNDERPINNING KNOWLEDGE ASSESSMENT</th></tr>
<tr><td>A person who has achieved this unit should be able to perform on job training to subordinates as per professional skills and job requirements.</td><td>1. Preparing training needs
2. Performing training of subordinates</td><td>The trainee should be able to:
• Select tools and equipment
• Prepare capability chart of the subordinates
• Conduct training needs assessment
• Identify knowledge and skills to be imparted
• Identify previous knowledge and skills possessed by the person to be trained</td><td>• A training program prepared to meet job requirements
• A person trained is able to execute tasks to required standards according to regulations</td><td>Knowledge evidence:
Detailed knowledge of:
Method used: The trainee should explain how to prepare training programme.

Principles: The trainee should explain the principles of carrying out training programme by using the four steps plan (prepare, present, try-out assign work).</td><td>This unit can be achieved at a work place or training institution.
The following equipment, tools and safety gears should be available:
• Laboratory
• Laboratory tools.
• Multimeter
• Laboratory equipment i.e.
- Distillers
- De-ionizers
- Spectrophotometer
- Gas chromatograph
- Incubators
- Oven
- Water bath
- Ultraviolet machine</td></tr>
</table>

PERFORMANCE CRITERIA	ELEMENTS	COMPETENCE ASSESSMENT			TRAINING REQUIREMENTS
		PROCESS ASSESSMENT	SERVICE ASSESSMENT	UNDERPINNING KNOWLEDGE ASSESSMENT	
		• Prepare a training programme for the subordinate • Carryout the training programme by using four steps plan (prepare, present, try-out, assign work) • Continually assess progress of the trainees • Make necessary adjustments to the training programme schedule • Clean the work area • Store tools, equipment, safety gears and other items		**Theories:** The trainee should explain: • Necessity of identifying previous knowledge and skill of the person to be trained • The importance of step by step guidance from simple to complex tasks **Circumstantial knowledge:** **Detailed knowledge about:** basic principles of educational psychology.	- Balances - Microscopes - Firefighting equipment - Meter bridge - Fume cupboards - Micrometer screw gauge - Vernier caliper - Thermometer - pH meter - Hydrometer • Computer • TV • Furnace • Crucible • Pneumatic and hydraulic pressure gauges • Surface block • Surface block • First aid kit • Firefighting equipment

PERFORMANCE CRITERIA	ELEMENTS	COMPETENCE ASSESSMENT			TRAINING REQUIREMENTS
		PROCESS ASSESSMENT	SERVICE ASSESSMENT	UNDERPINNING KNOWLEDGE ASSESSMENT	
					• Emergency exit • Overhead projector • Organization structure • Safety gears • Power hacksaw • Digital multimeter • Analogue meter • Oscilloscope

LEVEL III

MODULE NUMBER: 7 **MODULE TITLE: MANAGING COMPUTER ACTIVITIES**

UNIT NO: 5 **UNIT TITLE: PREPARING REPORTS** **DURATION: 15 HRS**

PERFORMANCE CRITERIA	**ELEMENTS**	**COMPETENCE ASSESSMENT**			**TRAINING REQUIREMENTS**
		PROCESS ASSESSMENT	**SERVICE ASSESSMENT**	**UNDERPINNING KNOWLEDGE ASSESSMENT**	
A person who has achieved this unit should be able to prepare reports as per management requirements.	1. Collecting Information 2. Submitting technical report	**The trainee should be able to:** • Select tools and equipment • Collect information • Write technical reports • Prepare budget report • Prepare action plan • Keep records	The prepared reports contain required contents as per management requirements.	**Knowledge evidence:** **Detailed knowledge of:** **Method used:** The trainee should explain: • Preparation of technical reports • Keeping of records **Principles:** The trainee should explain the principles of: • Supervision • Reporting	This unit can be achieved at a work place or training institution. The following equipment, tools, materials and safety gears should be available: • Office/table and chairs • Stationery • Computer • Printer • Job card • Subordinates reports • Binding machine • Photocopy machine • Overcoat • Safety boots

PERFORMANCE CRITERIA	ELEMENTS	COMPETENCE ASSESSMENT			TRAINING REQUIREMENTS
		PROCESS ASSESSMENT	SERVICE ASSESSMENT	UNDERPINNING KNOWLEDGE ASSESSMENT	
				Theories: The trainee should explain: • Importance of reports • Contents of reports • Steps of writing of technical reports • Types of reports and their vises **Circumstantial knowledge:** **Detailed knowledge about:** • Confidentiality while preparing reports • Safe handling of tools, equipment and documents • Waste disposal	

LEVEL III

MODULE NUMBER: 7 **MODULE TITLE: MANAGING COMPUTER ACTIVITIES**

UNIT NO: 6 **UNIT TITLE: MANAGING COMPUTER BUSINESS** **DURATION: 20 HRS**

PERFORMANCE CRITERIA	ELEMENTS	COMPETENCE ASSESSMENT			TRAINING REQUIREMENTS
		PROCESS ASSESSMENT	SERVICE ASSESSMENT	UNDERPINNING KNOWLEDGE ASSESSMENT	
A person who has achieved this unit should be able to manage laboratory business as per laboratory regulations.	1. Performing entrepreneurial tactics 2. Conducting manpower planning 3. Supervising Junior workers	**The trainee should be able to:** • Calculate total project cost • Prepare project write up • Select appropriate site for establishing business • Plan preventive maintenance • Plan machine loading • Plan delivery dates • Acquire land/building for setting the business • Purchase basic hand tools and equipment • Perform manpower planning	Managed laboratory business conforms to stipulated regulations.	**Knowledge evidence:** **Detailed knowledge of:** **Method used:** The trainee should explain how to: • Establish and run workshop business • Analyze profit and loss **Principles:** The trainee should explain principles used to: • Acquire capital from the Bank or an NGO • Calculate business profit and loss • Manage private business • Manage non private business **Theories:** The trainee should explain:	This unit can be achieved at a work place or training institution. The following equipment, tools and safety gears should be available: • Laboratory layout chart • Business films/video cassettes • Business magazines • Laboratory business regulations • Scheduled maintenance of tools and equipment • Job card sheets • Stationeries • Receipt book • Invoice books • Laboratory tools and equipment

PERFORMANCE CRITERIA	ELEMENTS	COMPETENCE ASSESSMENT			TRAINING REQUIREMENTS
		PROCESS ASSESSMENT	SERVICE ASSESSMENT	UNDERPINNING KNOWLEDGE ASSESSMENT	
		• Prepare at least six months salary for potential workers • Supervise provision of payment invoices and receipts • Identify labour and overhead costs • Analyze profit and loss • Prepare business brochures		• Meaning of terms business, entrepreneurship, entrepreneur, and enterprise • Project write up procedures • Good customer care **Circumstantial knowledge:** **Detailed knowledge about:** • Marketing and use of mass media • Safe handling of business capital • Waste proposal	• Personal computer • Laboratory store • Laboratory office • Tool ledger book • Safety gears • Machine manuals • Time cards